FHIR-STU3 — Security and Privacy Module (sections 6 – 6.4.11)

Source: HL7 FHIR STU3 (v3.0.2). Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

6 Security and Privacy Module

URL: https://hl7.org/fhir/STU3/secpriv-module.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group FHIR Infrastructure & Security | Ballot Status: Informative

6.0 Security and Privacy Module

6.0.1 Introduction

The Security and Privacy Module describes how to protect a FHIR server (through access control and authorization), how to document what permissions a user has granted (consent), and how to keep records about what events have been performed (audit logging and provenance). FHIR does not mandate a single technical approach to security and privacy; rather, the specification provides a set of building blocks that can be applied to create secure, private systems.

6.0.2 Index

The Security and Privacy module includes the following materials:

Resources | Datatypes | Implementation Guidance and Principles
Consent Provenance Audit Event | Signature | Security Principles Security Labels

6.0.3 Security

FHIR is focused on the data access methods and encoding leveraging existing Security solutions. Security in FHIR needs to focus on the set of considerations required to ensure that data can be discovered, accessed, or altered only in accordance with expectations and policies. Implementation should leverage existing security standards and implementations to ensure that:

* All communications can be encrypted to prevent unauthorized access.

* No information leaks when errors occur

* No active script content can be injected into narrative resources

* Full audit trails can be constructed and used to detect anomalous access patterns

For general security considerations and principles, see Security. Please leverage mature Security Frameworks cover device security, cloud security, big-data security, service-to-service security, etc. See NIST Mobile Device Security and OWASP Mobile Secuity . These security frameworks include prioritized lists of most important concerns.

6.0.4 Privacy

Privacy in FHIR includes the set of considerations required to ensure that individual data are treated according to an individual's Privacy Principles. FHIR includes implementation guidance to ensure that:

* Individual preferences can be communicated through standards-based protocols (e.g., OAuth, User-Managed Access) or using an explicit FHIR representation (Consent)

* Resources can be tagged to indicate the sensitivity or confidentiality of the data they represent (Security Labels)

* Data access records and audit logs can be shared with individuals, e.g. for accounting of disclosures (Audit Event)

6.0.5 Common Use Cases

6.0.5.1 Authorization and Access Control

Use case: A FHIR server should ensure that API access is allowed for authorized requests, and denied for unauthorized requests.

Approach: Authorization details can vary according to local policy, and according to the access scenario (e.g. sharing data among institution-internal subsystems vs. sharing data with trusted partners vs. sharing data with third-party user-facing apps). In general, FHIR enables a separation of concerns between the FHIR REST API and standards-based authorization protocols like OAuth. For the use case of user-facing third-party app authorization, we recommend the OAuth-based SMART protocol see Security: Authentication as an externally-reviewed authorization mechanism with a real-world deployment base — but we note that community efforts are underway to explore a variety of approaches to authorization. For further details, see Security: Authorization and Access Control.

6.0.5.2 Audit Logging

Use case: "A FHIR server should keep a complete, tamper-proof log of all API access and other security- and privacy-relevant events".

Approach: FHIR provides an AuditEvent resource suitable for use by FHIR clients and servers to record when a security or privacy relevant event has occurred. This form of audit logging records as much detail as reasonable at the time the event happened. The FHIR AuditEvent is aligned and cross-referenced with IHE Audit Trail and Node Authentication (ATNA) Profile. For details, see Security: Audit.

6.0.5.3 Privacy Consent

Use case: "Documentation of a Patient's Privacy Consent Directive - rules for Collection, Use, and Disclosure of their health data."

Approach: FHIR provides a Consent resource suitable for use by FHIR clients and servers to record current Privacy Consent state. The meaning of a consent or the absence of the consent is a local policy concern. The Privacy Consent may be a pointer to privacy rules documented elsewhere, such as a policy identifier or identifier in XACML. The Privacy Consent has the ability to point at a scanned image of an ink-on-paper signing ceremony, and supports digital signatures through use of Provenance. The Privacy Consent has the ability to include some simple FHIR centric base and exception rules.

All uses of FHIR Resources would be security/privacy relevant and thus should be recorded in an AuditEvent. Those access qualifying as a Disclosure should additionally be recorded as a Disclosure, see Disclosure Audit Event Example.

For Privacy Consent guidance and examples, see Consent Resource.

6.0.5.4 Provenance

Use case: "All FHIR Resources should be capable of having the Provenance fully described."

Approach: FHIR provides the Provenance resource suitable for use by FHIR clients and servers to record the full provenance details: who, what, where, when, and why. A Provenance resource can record details for Create, Update, and Delete; or any other activity. Generally Read operations would be recorded using AuditEvent. Many Resources include these elements within, this is done when that provenance element is critical to the use of that Resource. This is overlap is expected and cross-referenced on the W5 report. For details, see Provenance Resource.

6.0.5.5 Signature

Use case: "Digital Signature is needed to prove authenticity, integrity, and non-repudiation."

Approach: FHIR Resources are often parts of Medical Record, or are communicated as part of formal Medical Documentation. As such there is a need to cryptographically bind a signature so that the receiving or consuming actor can verify authenticity, integrity, and non-repudiation. This functionality is provided through the signature element in Provenance Resource. Where the signature can be any local policy agreed to signature including Digital Signature methods and Electronic Signature. For details, see Security: Digital Signatures.

6.0.6 Developmental Roadmap

In the STU3 release, FHIR includes building blocks and principles for creating secure, privacy-oriented health IT systems; FHIR does not mandate a single technical approach to security and privacy.

In future releases, we are anticipate including guidance on:

* incorporate the SMART on FHIR authorization specification, for user-authorized apps,

* methods for organization-to-organization authorization,

* more details about how to use digital Signatures for data integrity and non-repudiation, including an approach that supports some level of manipulation of resources (e.g. separating the entries in a bundle, or conversion between XML and JSON during processing),

* more detailed Consent management, including support for specific consent use cases.

6.1 FHIR Security

URL: https://hl7.org/fhir/STU3/security.html

The FHIR RESTful API provides a number of ways that a client may request or create information. When designing a system to authorize access to information, all potential access methods must be considered. They include the following:

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 4 | Ballot Status: Trial Use

Fast Healthcare Interoperability Resources (FHIR) is not a security protocol, nor does it define any security related functionality. However, FHIR does define exchange protocols and content models that need to be used with various security protocols defined elsewhere. This section gathers all information about security in one section. A summary:

* Time Keeping - all clocks should be synchronised using NTP/SNTP, and the design of the system should be robust against a system clock with the wrong value

* Communications Security - all exchange of production data should be secured using TLS/SSL (e.g. https)

* Authentication - Users/Clients may be authenticated in any way desired. For web-centric use, OAuth is recommended

* Authorization/Access Control - FHIR defines a Security Label infrastructure to support access control management. FHIR may also define a set of resources to administer access control management, but does not define any at present

* Audit - FHIR defines provenance and audit event resources suitable for tracking the origins, authorship, history, status, and access of resources

* Digital Signatures - FHIR includes several specifically reserved locations for digital signatures

* Attachments - FHIR allows for binary resources and attachments. These have their own concerns

* Labels - FHIR allows for set of security related tags that affect that way resources are handled

* Data Management Policies - FHIR defines a set of capabilities to support data exchange. Not all the capabilities that FHIR enables may be appropriate or legal for use in some combinations of context and jurisdiction (e.g. HIPAA for exchange between institutions). It is the responsibility of implementers to ensure that relevant regulations and other requirements are met

* Narrative - Care must be taken when displaying the narrative from FHIR resources

Time critical concerns regarding security flaws in the FHIR specification should be addressed to the FHIR email list for prompt consideration.

Implementers should track the developing IHE IUA Profile for additional security considerations.

6.1.0.1 General Considerations

A production FHIR system will need some kind of security sub-system that administers users, user authentication, and user authorization. Where this subsystem fits into the deployment architecture is a matter for system design:

| | The consumer that is using a healthcare related system The client application the user is using (application, mobile app, website, etc.) The security system (authentication and access control) The clinical/healthcare repository
| The consumer that is using a healthcare related system
| The client application the user is using (application, mobile app, website, etc.)
| The security system (authentication and access control)
| The clinical/healthcare repository

In this diagram, the red lines represent FHIR interfaces. From the perspective of the FHIR API, the client (consumer of FHIR services) may either interact with a security system that manifests as a FHIR server, and which depends on a subsequent FHIR interface to provide the actual storage, or either the client or server interacts with the security system independently. In each of these 3 scenarios, the different components may be assembled into applications or network components differently, but the same logical layout applies. The FHIR specification assumes that a security system exists, and that it may be deployed in front of or behind the FHIR API.

The security system includes the following subsystems:

* Authentication: identifies and authenticates the user

* Access Control decision engine: decides whether FHIR operations are allowed

* Audit Log: records actions to allow for subsequent review and detection of intrusion or inappropriate usage

Because there are a plethora of standards relating to the administration and functionality of the security system, FHIR does not provide user, profile, or other such administration resources. Instead, the FHIR resources are the targets of the policies expressed in these other approaches. What FHIR does specify is a way to apply security labels to resources so that a security system may use these (along with the contents of the resources if appropriate) to determine whether a user is authorized to perform a particular FHIR operation or not.

6.1.0.2 Communications

For the RESTful API, normal HTTP security rules apply. Please follow the HTTP specification Security Considerations section 15 . The Service Base URL will specify whether SSL is required. Client authentication may be required by the server, possibly including the requirement for client certificates.

TLS/SSL SHOULD be used for all production data exchange. The TLS/SSL communications are established prior to any HTTP command/response, so the whole FHIR interaction is protected by the SSL/TLS communications. The security of the endpoints of the TLS/SSL communications must be risk-managed, so as to prevent inappropriate risks (e.g. audit logging of the GET parameters into an unprotected audit log).

To support browser-based client applications, servers SHOULD implement cross-origin resource sharing for the REST operations.

6.1.0.3 Authentication

Other than testing systems, FHIR servers should authenticate the clients. The server may choose to authenticate the client system and trust it, or to authenticate the individual user by a variety of techniques. For web-centric use, OpenID Connect may be used to authenticate users and OAuth may be used to authenticate and/or authorize the users. The Smart-On-FHIR profile on OAuth is tightly integrated with FHIR and is the preferred method for using OAuth.

The HEART Working Group has developed a set of privacy and security specifications that enable an individual to control the authorization of access to RESTful health-related data sharing APIs, and to facilitate the development of interoperable implementations of these specifications by others.

6.1.0.4 Authorization/Access Control

Correctly identifying people, devices, locations and organizations is one of the foundations that any security system is built on. Most applications of security protocols, whether authentication, access control, digital signatures, etc. rely on the correct mapping between the relevant resources and the underlying systems. Note that this isn't necessary. There is nothing in FHIR that requires or relies on any security being in place, or any particular implementation. However, real world usage will generally require this.

A holder of data should not allow the data to be communicated unless there are sufficient assurances that the other party is authorized to receive it. This is true for a client creating a resource through a PUT/POST, as much as it is true for a server returning resources on a GET. The presumption is that without proper authorization, to the satisfaction of the data holder, the data does not get communicated.

Two of the classic Access Control models are: Role-Based Access Control (RBAC), and Attribute-Based Access Control (ABAC).

In Role-Based Access Control (RBAC), permissions are operations on an object that a user wishes to access. Permissions are grouped into roles. A role characterizes the functions a user is allowed to perform. Roles are assigned to users. If the user's role has the appropriate permissions to access an object, then that user is granted access to the object. FHIR readily enables RBAC, as FHIR Resources are object types and the CRUDE (Create, Read, Update, Delete, Execute) events (the FHIR equivalent to permissions in the RBAC scheme) are operations on those objects.

In Attribute-Based Access Control (ABAC), a user requests to perform operations on objects. That user's access request is granted or denied based on a set of access control policies that are specified in terms of attributes and conditions. FHIR readily enables ABAC, as instances of a Resource in FHIR (again, Resources are object types) can have attributes associated with them. These attributes include security tags, environment conditions, and a host of user and object characteristics, which are the same attributes as those used in ABAC. Attributes help define the access control policies that determine the operations a user may perform on a Resource (in FHIR) or object (in ABAC). For example, a tag (or attribute) may specify that the identified Resource (object) is not to be further disclosed without explicit consent from the patient.

The rules behind the access control decision are often very complex, and potentially depend on information sourced from:

* Client, such as user identity, user role, location, level of assurance

* Resource, such as confidentiality, sensitivity, type of data, date ranges covered by the data, author of the data

* Patient, such as the patient identity, patient relationship to the user, patient consent policies

* Context of the transaction, system identity, time-of-day, purpose of use, workflow state, and transport security

For one source of further information, see the IHE Access Control white paper

Access control constraints may result in data returned in a read or search being redacted or otherwise restricted. See Variations between Submitted data and Retrieved data.

6.1.0.4.1 Access Control Considerations

* The basic CRUD methods on resources. A security implementation must evaluate whether a client is allowed to read, update create or delete a given resource.

* Chained search provides the ability to disclose information on related resources. A security implementation must consider whether a client has the permission to access the resource being searched on, as well as the chained resource(s)

* _include and _revinclude search parameters allow client to request related resources. A security implementation must determine if the client has access to the included resources.

* security labels

* Several resources, including Bundle, Composition, Group and List, are designed to contain other resources. A security implementation should consider whether access to an individual resource, such as a Bundle, should permit access to all resources contained within the resource.

* FHIR defines several operations that may be supported by a server. Security implementations must evaluate whether a client has the ability to invoke these operations and what information should be returned from them. Fetch Encounter Record, Evaluate Measure, Observation Statistics, Find patient matches using MPI based logic and Fetch Patient Record specifically provide the ability to disclose patient information.

* Batch and transaction processing provide ways for clients to create and update information in bulk. Security implementations should consider whether a client has the ability to initiate one of these interactions and make authorization decisions on each action in the batch/transaction.

* Security implementations must be aware of the Break the Glass protocol (e.g. break the glass) (example).

6.1.0.4.2 Access Denied Response Handling

A web-server, especially hosting FHIR, must choose the response carefully when an Access Denied condition exists. Returning too much information may expose details that should not be communicated. The Access Denied condition might be because of missing but required Authentication, the user is not authorized to access the endpoint, the user is not authorized to access specific data, or other policy reasons.

To balance usability of the returned result vs appropriate protection, the actual result method used needs to be controlled by policy and context. Typical methods of handling Access Denied used are:

Return a Success with Bundle containing zero results – This result is indistinguishable from the case where no data is known. When consistently returned on Access Denied, this will not expose which patients exist, or what data might be blinded. This

method is also consistent with cases where some results are authorized while other results are blinded. This can only be used when the returning a Bundle is a valid result.

Return a 404 “Not Found” – This also protects from data leakage as it is indistinguishable from a query against a resource that doesn’t exist. It does however leak that the user authentication is validated.

Return a 403 “Forbidden” – This communicates that the reason for the failure is an Authorization failure. It should only be used when the client and/or user is well enough known to be given this information. Thus this method is most used when the user is allowed to know that they are forbidden access. It doesn’t explain how the user might change things to become authorized.

Return a 401 “Unauthorized” – This communicates that user authentication was attempted and failed to be authenticated.

6.1.0.5 Audit Logging

FHIR provides an AuditEvent resource suitable for use by FHIR clients and servers to record when a security or privacy relevant event has occurred. This form of audit logging records as much detail as reasonable at the time the event happened.

When used to record security and privacy relevant events, the AuditEvent can then be used by properly authorized applications to support audit reporting, alerting, filtering, and forwarding. This model has been developed and used in healthcare for a decade as IHE-ATNA profile . ATNA log events can be automatically converted to AuditEvent resources, and from there, client applications are able to search the audit events, or subscribe to them.

With regard to HTTP logs, implementers need to consider the implications of distributing access to the logs. HTTP logs, including those that only contain the URL itself, should be regarded as being as sensitive as the resources themselves. Even if direct PHI is kept out of the logs by careful avoidance of search parameters (e.g. by using POST), the logs will still contain a rich set of information about the clinical records.

6.1.0.6 Digital Signatures

This specification recommends the use of W3C Digital Signatures for signatures. Resources can be signed using the Provenance resource to carry a detached digital signature . The Signature datatype is available to support various signature types including non-repudiation purposes. Further details on creation and validation of Signatures are defined.

In addition, documents may be signed using an enveloped signature. A specification for enveloped signature is profiled in the IHE DSG profile .

Neither of these definitions prohibits the use of other ways of using digital signatures.

STU Note: the use of signatures with RESTful interfaces is a poorly understood area, and we would welcome reports of implementation experience. See discussion on use of Digital Signature in FHIR

Feedback is welcome here .

6.1.0.7 Attachments

Several FHIR resources include attachments. Attachments can either be references to content found elsewhere or included inline encoded in base64. Attachments represent security risks in a way that FHIR resources do not, since some attachments contain executable code. Implementers should always use caution when handling resources.

6.1.0.8 Security Labels

See Security Labels.

6.1.0.9 Narrative

FHIR resources include an XHTML narrative, so that applications can display the contents of the resource to users without having to fully and correctly process the data in the resource. However, displaying HTML is associated with several known security issues that have been observed in production systems in other contexts (e.g. with CDA). For this reason, the FHIR narrative is not allowed to contain active content. However, care is still needed when displaying the narrative:

* Validate the narrative (the standard FHIR schemas do not allow active content, and the reference implementations won't handle it). Note, though, that external references could still be included in CSS, and these are outside the scope of schemas and reference implementations.

* Ensure that any external references to images or anchors (e.g. outside the resource) do not cause the display software to leak sensitive information in headers

* Do not allow external links to run in a privileged context such as the EHR unless you are sure they can be trusted

* Care should be taken to differentiate HTTP RESTful (API) from browser based server content. Specifically, one should separate user session cookies, as an attacker could create content that serves up with content-type "text/html" and has content like "<script>send_to_attacker(document.cookie);</script>".

Also note that the inclusion of an external reference to an image can allow the server that hosts the image to track when the resource is displayed. This may be a feature or a problem depending on the context.

In addition to narrative, Documents may also contain stylesheets. Unlike with CDA, the stylesheets are simple CSS stylesheets, not executable XSLT, so the same security risks do not apply. However CSS stylesheets may still reference external content (e.g. background images), and applications displaying documents should ensure that CSS links are not automatically followed without checking their safety first, and that session/identifying information does not leak with any use of

external links.

6.1.1 Security Labels

URL: https://hl7.org/fhir/STU3/security-labels.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 3 | Ballot Status: Trial Use

A security label is a concept attached to a resource or bundle that provides specific security metadata about the information it is fixed to. The Access Control decision engine uses the security label together with any provenance resources associated with the resource and other metadata (e.g. the resource type, resource contents, etc.) to

* approve read, change, and other operations

* determine what resources can be returned

* determine what handling caveats must be conveyed with the data

Security Labels enable more data to flow as they enable policy fragments to accompany the resource data.

The intent of a security label is that the recipient of resources or bundles with security-tags is obligated to enforce the handling caveats of the tags and carry the security labels forward as appropriate.

Security labels are only a device to connect specific resources, bundles, or operations to a wider security framework; a full set of policy and consent statements and their consequent obligations is needed to give the labels meaning. As a consequence of this, security labels are most effective in fully trusted environments - that is, where all trading partners have agreed to abide by them in a Mutual Trust Framework. Note also that security labels support policy, and specific tagging of individual resources is not always required to implement policy correctly.

In the absence of this kind of pre-agreement, Security Labels may still be used by individual parties to assist with security role checking, but they may not all be recognized and enforced, which in turn limits what information is allowed to flow.

Local agreements and implementation profiles for the use security labels should describe how the security labels connect to the relevant consent and policy statements, and in particular:

* Which Security Labels are able to be used

* What to do if a resource has an unrecognized security label on it

* Authoring obligations around security labels

* Operational implications of security labels

This specification defines a basic set of labels for the most common use cases trading partners, and also a wider array of security labels that allow much finer grained management of the information.

6.1.1.1 Representing Security Labels

A security label is represented as a Coding, with the following important properties:

system | The coding scheme from which label is taken (see code system URI, and below)
code | a code from the coding scheme that identifies the security label and code is an value from the code system
display | The display form for the code (mostly for use when a system doesn't recognize the code)

An example XML Patient Resource with a "Restricted" tag associated with it, as represented in an HTTP response:

```
<Patient xmlns="http://hl7.org/fhir">
  <meta>
    <security>
      <system value="http://hl7.org/fhir/v3/Confidentiality"/>
      <code value="R"/>
      <display value="Restricted"/>
    </security>
  </meta>
...  [snip] ...
</Patient>
```

A JSON search result that includes a resource that the receiving application must delete all copies of the resource after using it:

```
{
  "resourceType" : "Bundle",
  "type" : "searchset",
  ... other headers etc.....
  "entry" : [
     ... other entries ....
     {
       "resource": {
         "id" : "1",
         "meta" : {
           "security" : [{
             "system" : "http://hl7.org/fhir/v3/ActCode",
             "code" : "DELAU",
             "display" : "delete after use"
           }]
         }
         ... other content etc.....
```

```
      }
    },
    ... other entries ....
  ]
}
```

Note: the actual terms used in these examples are described below.

The basic framework for security labels is described by the HL7 Healthcare Classification System (HCS; ref todo). This specification identifies how security labels are defined and provides a relatively comprehensive list of labels. All of the HCS defined labels (see below for the lists) can be used as security labels on FHIR resources and bundles (e.g. requests and responses).

In addition, other security labels not defined here or in the HCS can be defined by jurisdictions, vendors and/or projects and used as appropriate. However, note that:

* Defining additional security labels will increase costs associated with information and system portability

* Implementation guides and applications SHOULD always use the applicable label defined by the HCS if one exists

Note: The use of security labels and the expression of common shared security policies is a matter of ongoing discussion and development in several communities at this time.

6.1.1.2 Core Security Labels

This specification defines a set of core security labels for all FHIR systems. All conformant FHIR Applications SHOULD use these labels where appropriate. For all of these labels, how they are operationalised - their use and interpretation - is subject to the applicable Mutual Trust Framework agreement as described above.

Name/ Tag | Description

Context of Use

Purpose Of Use | These Purpose Of Use (system = http://hl7.org/fhir/v3/PurposeOfUse) is an indication of a reason for performing one or more operations on information. which may be permitted by source system's security policy in accordance with one or more privacy policies and consent directives. Such as collecting personal health information for research or public health purposes. Notes may be used as: The rationale or purpose for a request for data The use limitation on a data Bundle See discussion on HCS below

Data Sensitivity

Confidentiality codes | These confidentiality class (system = http://hl7.org/fhir/v3/Confidentiality) can be applied to any resource or bundle. They are generally assigned by the author of the resource, but can be modified subsequently as a matter of operational management. The Confidentiality classifications describe the sensitivity of the information in a resource with regard to whether it should made available or disclosed to unauthorized individuals, entities, or processes. Notes: In the absence of a confidentiality code, the basic confidentiality of a resource may be implied by its definition and content; e.g. a patient's condition is far more likely to be confidential than a practitioner resource, and a Diagnostic Report with an HIV test is always highly confidential, whereas a routine electrolytes is rarely particularly confidential The confidentiality of a bundle is always as confidential as the most confidential resource in the bundle The additional security lables are more specific to support very specific fine-grained access control, and should always be used in association with an appropriate confidentiality label. See discussion on HCS below

Control of Flow

Delete After Use: ActCode.DELAU | An application receiving a resource with this label must delete all copies after the immediate use for which the data was exchanged, is complete. Notes: This may imply a prohibition not storing the resource in any audit trail as well Additional security labels are allowed to make exceptions to the blanket restriction this implies. This allows a resource to be exchanged with a blanket rule not to retain copies unless the exact rules for retaining it can be followed

Do Not Re-use: ActCode.NOREUSE | An application receiving a resource with this label may only use it for the immediate purpose of use. In particular, the application is not authorized to re-distribute (i.e. exchange this resource with any other application). Notes: The exact interpretation of "immediate purpose of use" and the boundaries of "the application" are determined by local policy Additional security labels are allowed to make exceptions to the blanket restriction this implies. This allows a resource to be exchanged with a blanket rule not to re-use unless the exact rules for doing so can be followed

6.1.1.3 Break The Glass

There is a special security label to support the commonly encountered "break-the-glass" protocol, where a clinician (usually in an emergency context) requests emergency unauthorized access to the patient's record. This specification does not make any policy recommendations or rules about the operation, merely provides support for it. See this paper for discussion of the issues involved in break-the-glass operations.

When the operation occurs, it is represented as a security label on the request, rather than on a resource, and so is represented differently. The break the glass tag needs to be used as part of an agreed policy and protocol. FHIR does not attempt to define this policy or protocol, it must be agreed on an implementation by implementation basis. For example as a URL:

Break The Glass | http://hl7.org/fhir/security-label#break-the-glass | The requester is asking for emergency access for patient treatment. Typically, this means that the patient is unconscious and not able to provide relevant information or consent.

The URL is represented in the request as a web category :

```
HTTP/1.1 GET fhir/Patient/482735/condition
Content-Type: text/xml
Access-Control-Allow-Origin: *
Last-Modified: Thu, 19 Nov 2013 07:07:32 +1100
ETag: 24
Category: http://hl7.org/fhir/security-label#break-the-glass; scheme="http://hl7.org/fhir/tag/security"; label="Break
    The Glass"
```

6.1.1.4 Healthcare Privacy and Security Classification System (HCS)

The security labels described above are a subset of the full set of security labels defined by the HL7 Healthcare Privacy and Security Classification System . The HCS defines 5 categories of security labels that may be applied to a resource:

Security Label | Card. | Values | Description

Confidentiality Classification | 0..1 | ConfidentialityClassification | Security label metadata classifying an IT resource (clinical fact, data, information object, service, or system capability) according to its level of sensitivity, which is based on an analysis of applicable privacy policies and the risk of financial, reputational, or other harm to an individual or entity that could result if made available or disclosed to unauthorized individuals, entities, or processes. Example Uses: Unrestricted, Normal, Very restricted

Sensitivity Category | 0..* | InformationSensitivityPolicy | Security label metadata that "segments" an IT resource by categorizing the value, importance, and vulnerability of an IT resource perceived as undesirable to share. Example Uses: STDs, Psychiatric care, Celebrity status

Compartment Category | 0..* | Compartment | Security label metadata that "segments" an IT resource by indicating that access and use is restricted to members of a defined community or project Note: this is a different use of "Compartment" to the Patient Compartment use. Example Uses: Research, HR records

Integrity Category | 0..* | SecurityIntegrityObservationValue | Security label metadata that "segments" an IT resource by conveying the completeness, veracity, reliability, trustworthiness, and provenance of an IT resource Example Uses: Anonymized, signed, patient reported

Handling Caveat | 0..* | SecurityControlObservationValue | Security label metadata conveying dissemination controls and information handling instructions such as obligations and retention policies to which an IT resource custodian or receiver must comply. This type of handling caveat SHALL be assigned to a clinical fact if required by jurisdictional or organizational policy, which may be triggered by a patient consent directive Example Uses: do not disclose, various restrictions on use, and policy marks

Each of these security labels identifies a ValueSet that lists a set of possible codes for the security label.

6.1.1.4.1 Jurisdiction Specific Security Labels

The HL7 Healthcare Classification System also allows for Realm-specific privacy law or policy category codes for use in security labels in particular domains. These domains are included with this specification:

Security Label | Card. | Values | Description

US Privacy Law | 0..* | ActUSPrivacyLaw | Security label metadata that "segments" an IT resource by indicating the legal provisions to which the assignment of a Confidentiality Classification complies in the US.

6.1.2 Digital Signatures

URL: https://hl7.org/fhir/STU3/signatures.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Trial Use

todo - digital signatures in FHIR.

6.2 Resource Consent - Content

URL: https://hl7.org/fhir/STU3/consent.html

element.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Community Based Collaborative Care Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

A record of a healthcare consumer’s policy choices, which permits or denies identified recipient(s) or recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of time.

6.2.1 Scope and Usage

The purpose of this Resource is to be used to express a Consent regarding Healthcare. There are four anticipated uses for the Consent Resource, all of which are written or verbal agreements by a healthcare consumer [grantor] or a personal representative, made to an authorized entity [grantee] concerning authorized or restricted actions with any limitations on purpose of use, and handling instructions to which the authorized entity must comply:

* Privacy Consent Directive: Agreement to collect, access, use or disclose (share) information.

* Medical Treatment Consent Directive: Consent to undergo a specific treatment (or record of refusal to consent).

* Research Consent Directive: Consent to participate in research protocol and information sharing required.

* Advance Care Directives: Consent to instructions for potentially needed medical treatment (e.g. DNR).

This resource is scoped to cover all four uses, but at this time, only the privacy use case is modeled. The scope of the resource may change when the other possible scopes are investigated, tested, or profiled.

A FHIR Consent Directive instance is considered the encoded legally binding Consent Directive if it meets requirements of a policy domain requirements for an enforceable contract. In some domains, electronic signatures of one or both of the parties to the content of an encoded representation of a Consent Form is deemed to constitute a legally binding Consent Directive. Some domains accept a notary’s electronic signature over the wet or electronic signature of a party to the Consent Directive as the additional identity proofing required to make an encoded Consent Directive legally binding. Other domains may only accept a wet signature, or may not require the parties’ signatures at all.

Whatever the criteria are for making an encoded FHIR Consent Directive legally binding, anything less than a legally binding representation of a Consent Directive must be identified as such, i.e., as a derivative of the legally binding Consent Directive, which has specific usage in Consent Directive workflow management.

Definitions:

Consent | The record of a healthcare consumer’s policy choices, which permits or denies identified recipient(s) or recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of time

Consent Directive | The legal record of a healthcare consumer's agreement with a party responsible for enforcing the consumer’s choices, which permits or denies identified actors or roles to perform actions affecting the consumer within a given context for specific purposes and periods of time

Consent Form | Human readable consent content describing one or more actions impacting the grantor for which the grantee would be authorized or prohibited from performing. It includes the terms, rules, and conditions pertaining to the authorization or restrictions, such as effective time, applicability or scope, purposes of use, obligations and prohibitions to which the grantee must comply. Once a Consent Form is “executed” by means required by policy, such as verbal agreement, wet signature, or electronic/digital signature, it becomes a legally binding Consent Directive.

Consent Directive Derivative | Consent Content that conveys the minimal set of information needed to manage Consent Directive workflow, including providing Consent Directive content sufficient to: Represent a Consent Directive Register or index a Consent Directive Query and respond about a Consent Directive Retrieve a Consent Directive Notify authorized entities about Consent Directive status changes Determine entities authorized to collect, access, use or disclose information about the Consent Directive or about the information governed by the Consent Directive. Derived Consent content includes the Security Labels encoding the applicable privacy and security policies. Consent Security Labels inform recipients about specific access control measures required for compliance.

Consent Statement | A Consent Directive derivative has less than full fidelity to the legally binding Consent Directive from which it was "transcribed". It provides recipients with the full content representation they may require for compliance purposes, and typically include a reference to or an attached unstructured representation for recipients needing an exact copy of the legal agreement.

Consent Registration | The legal record of a healthcare consumer's agreement with a party responsible for enforcing the consumer’s choices, which permits or denies identified actors or roles to perform actions affecting the consumer within a given context for specific purposes and periods of timeA Consent Directive derivative that conveys the minimal set of information needed to register an active and revoked Consent Directive, or to update Consent status as it changes during its lifecycle.

Consent Query/Response Types | The FHIR Consent Resource specifies multiple Consent Search parameters, which support many types of queries for Consent Resource content. There are several Query/Response patterns that are typically used for obtaining information about consent directive content for the following use cases: Find Active Consent Directive: A query that includes sufficient consent directive content to determine whether a specific party is authorized to share information governed by a consent directive with another specific party. The Response is either: “Yes” meaning that both parties are authorized to share the information with one another. “No” meaning that the authorized querier is not permitted to share with another specific party “No information found” meaning that there is no active Consent Directive in which the querier is authorized to share the governed information. Find Consent Directive Authorized Entities: A query that includes sufficient consent directive content to return a list of entities with which the querier is authorized to share governed information. The response to an authorized querier is the list of any authorized entities with which the querier is permitted to share governed information. The response to an unauthorized querier is that “no information is found”. Find Consent Directive(s): A query that includes sufficient consent directive content to return a list of Consent Directive metadata for an authorized querier to determine what Consent Directives are available, and to locate and retrieve one or more of those Consent Directives as needed.

Policy context | Any organizational or jurisdictional policies, which may limit the consumer’s policy choices, and which includes the named range of actions allowed

Healthcare Consumer | The individual establishing his/her personal consent (i.e. Consenter). In FHIR, this is referred to as the 'Patient' though this word is not used across all contexts of care

6.2.1.1 Privacy Consent Directive (PCD)

Privacy policies define how Individually Identifiable Health Information (IIHI) is to be collected, accessed, used and disclosed. A Privacy Consent Directive as a legal record of a patient's (e.g. a healthcare consumer) agreement with a party responsible for enforcing the patient's choices, which permits or denies identified actors or roles to perform actions affecting the patient within a given context for specific purposes and periods of time. All consent directives have a policy context, which is any set of organizational or jurisdictional policies which may limit the consumer’s policy choices, and which include a named range of actions allowed. In addition, Privacy Consent Directives provide the ability for a healthcare consumer to delegate authority to a Substitute Decision Maker who may act on behalf of that individual. Alternatively, a consumer may author/publish their privacy preferences as a self-declared Privacy Consent Directive.

The Consent resource on FHIR provides support for alternative representations for expressing interoperable health information privacy consent directives in a standard form for the exchange and enforcement by sending, intermediating, or receiving systems of privacy policies that can be enforced by consuming systems (e.g., scanned documents, of computable structured entries elements, FHIR structures with optional attached, or referenced unstructured representations.) It may be used to represent the Privacy Consent Directive itself, a Consent Statement, which electronically represents a Consent Directive, or Consent Metadata, which is the minimum necessary consent content derived from a Consent Directive for use in workflow management.

6.2.2 Boundaries and Relationships

Consent management - particularly privacy consent - is complicated by the fact that consent to share is often itself necessary to protect. The need to protect the privacy of the privacy statement itself competes with the execution of the consent statement. For this reason, it is common to deal with 'consent statements' that are only partial representations of the full consent statement that the patient provided.

For this reason, the consent resource contains two elements that refer back to the source: a master identifier, and a direct reference to content from which this Consent Statement was derived. That reference can be one of several things:

* A reference to another consent resource from which this limited statement was derived

* A reference to a document format for the original source (e.g. PDF or CDA - see the HL7 CDAR2 ConsentDirective Implementation Guide , which incorporated the IHE Basic Patient Privacy Consents (BPPC)), either directly, or in a reference

* The source can be included in the consent as an attachment

The consent statements represent a chain that refers back to the original source consent directive. Applications may be able to follow the chain back to the source, but should not generally assume that they are authorized to do this.

Consent Directives are executed by verbal acknowledge or by being signed - either on paper, or digitally. Consent Signatures will be found in the Provenance resource (example consent and signature). Implementation Guides will generally make rules about what signatures are required, and how they are to be shared and used.

6.2.3 Background and Context

The Consent resource is structured with a base policy which is either opt-in or opt-out, followed by a listing of exceptions to that policy. The exceptions can be additional positive or negative exceptions upon the base policy. The set of exceptions include a list of data objects, list of authors, list of recipients, list of Organizations, list of purposeOfUse, and Date Range.

The enforcement of the Privacy Consent Directive is not included, but is expected that enforcement can be done using a mix of the various Access Control enforcement methodologies (e.g. OAuth, UMA, XACML). This enforcement includes the details of the enforcement meaning of the elements of the Privacy Consent Directive, such as the rules in place when there is an opt-in consent would be specific about which organizational roles have access to what kinds of resources (e.g. RBAC, ABAC). The specification of these details are not in scope for the Consent resource.

This resource is referenced by researchsubject

6.2.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Consent | I | | DomainResource | A healthcare consumer's policy choices to permits or denies recipients or roles to perform actions for specific purposes and periods of time + Either a Policy or PolicyRule Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Identifier for this record (external references)
status | ?!? | 1..1 | code | draft | proposed | active | rejected | inactive | entered-in-error ConsentState (Required)
category | ? | 0..* | CodeableConcept | Classification of the consent statement - for indexing/retrieval Consent Category Codes (Example)
patient | ? | 1..1 | Reference(Patient) | Who the consent applies to
period | ? | 0..1 | Period | Period that this consent applies
dateTime | ? | 0..1 | dateTime | When this Consent was created or indexed
consentingParty | ? | 0..* | Reference(Organization | Patient | Practitioner | RelatedPerson) | Who is agreeing to the policy and exceptions
actor | ? | 0..* | BackboneElement | Who|what controlled by this consent (or group, by role)
role | | 1..1 | CodeableConcept | How the actor is involved SecurityRoleType (Extensible)
reference | | 1..1 | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson) | Resource for the actor (or group, by role)
action | ? | 0..* | CodeableConcept | Actions controlled by this consent Consent Action Codes (Example)
organization | ? | 0..* | Reference(Organization) | Custodian of the consent
source[x] | ? | 0..1 | | Source from which this consent is taken
sourceAttachment | | | Attachment
sourceIdentifier | | | Identifier
sourceReference | | | Reference(Consent | DocumentReference | Contract | QuestionnaireResponse)
policy | | 0..* | BackboneElement | Policies covered by this consent
authority | I | 0..1 | uri | Enforcement source for policy
uri | I | 0..1 | uri | Specific policy covered by this consent
policyRule | ?I | 0..1 | uri | Policy that this consents to
securityLabel | ? | 0..* | Coding | Security Labels that define affected resources All Security Labels (Extensible)
purpose | ? | 0..* | Coding | Context of activities for which the agreement is made PurposeOfUse (Extensible)
dataPeriod | ? | 0..1 | Period | Timeframe for data controlled by this consent
data | ? | 0..* | BackboneElement | Data controlled by this consent
meaning | ? | 1..1 | code | instance | related | dependents | authoredby ConsentDataMeaning (Required)
reference | ? | 1..1 | Reference(Any) | The actual data reference
except | ? | 0..* | BackboneElement | Additional rule - addition or removal of permissions
type | ? | 1..1 | code | deny | permit ConsentExceptType (Required)
period | ? | 0..1 | Period | Timeframe for this exception
actor | ? | 0..* | BackboneElement | Who|what controlled by this exception (or group, by role)
role | | 1..1 | CodeableConcept | How the actor is involved SecurityRoleType (Extensible)
reference | | 1..1 | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson) | Resource for the actor (or group, by role)
action | ? | 0..* | CodeableConcept | Actions controlled by this exception Consent Action Codes (Example)
securityLabel | ? | 0..* | Coding | Security Labels that define affected resources All Security Labels (Extensible)
purpose | ? | 0..* | Coding | Context of activities covered by this exception PurposeOfUse (Extensible)
class | ? | 0..* | Coding | e.g. Resource Type, Profile, or CDA etc Consent Content Class (Extensible)
code | ? | 0..* | Coding | e.g. LOINC or SNOMED CT code, etc in the content Consent Content Codes (Example)

dataPeriod | ? | 0..1 | Period | Timeframe for data controlled by this exception
data | ? | 0..* | BackboneElement | Data controlled by this exception
meaning | ? | 1..1 | code | instance | related | dependents | authoredby ConsentDataMeaning (Required)
reference | ? | 1..1 | Reference(Any) | The actual data reference
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Consent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Identifier for this record (external references) --></identifier>
 <status value="[code]"/><!-- 1..1 draft | proposed | active | rejected | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept Classification of the consent statement - for indexing/retrieval --></category>
 <patient><!-- 1..1 Reference(Patient) Who the consent applies to --></patient>
 <period><!-- 0..1 Period Period that this consent applies --></period>
 <dateTime value="[dateTime]"/><!-- 0..1 When this Consent was created or indexed -->
 <consentingParty><!-- 0..* Reference(Organization|Patient|Practitioner|
   RelatedPerson) Who is agreeing to the policy and exceptions --></consentingParty>
 <actor>  <!-- 0..* Who|what controlled by this consent (or group, by role) -->
  <role><!-- 1..1 CodeableConcept How the actor is involved --></role>
  <reference><!-- 1..1 Reference(Device|Group|CareTeam|Organization|Patient|
    Practitioner|RelatedPerson) Resource for the actor (or group, by role) --></reference>
 </actor>
 <action><!-- 0..* CodeableConcept Actions controlled by this consent --></action>
 <organization><!-- 0..* Reference(Organization) Custodian of the consent --></organization>
 <source[x]><!-- 0..1 Attachment|Identifier|Reference(Consent|DocumentReference|
   Contract|QuestionnaireResponse) Source from which this consent is taken --></source[x]>
 <policy>  <!-- 0..* Policies covered by this consent -->
  <authority value="[uri]"/><!--  0..1 Enforcement source for policy -->
  <uri value="[uri]"/><!--  0..1 Specific policy covered by this consent -->
 </policy>
 <policyRule value="[uri]"/><!--  0..1 Policy that this consents to -->
 <securityLabel><!-- 0..* Coding Security Labels that define affected resources --></securityLabel>
 <purpose><!-- 0..* Coding Context of activities for which the agreement is made --></purpose>
 <dataPeriod><!-- 0..1 Period Timeframe for data controlled by this consent --></dataPeriod>
 <data>  <!-- 0..* Data controlled by this consent -->
  <meaning value="[code]"/><!-- 1..1 instance | related | dependents | authoredby -->
  <reference><!-- 1..1 Reference(Any) The actual data reference --></reference>
 </data>
 <except>  <!-- 0..* Additional rule -  addition or removal of permissions -->
  <type value="[code]"/><!-- 1..1 deny | permit -->
  <period><!-- 0..1 Period Timeframe for this exception --></period>
  <actor>  <!-- 0..* Who|what controlled by this exception (or group, by role) -->
   <role><!-- 1..1 CodeableConcept How the actor is involved --></role>
   <reference><!-- 1..1 Reference(Device|Group|CareTeam|Organization|Patient|
     Practitioner|RelatedPerson) Resource for the actor (or group, by role) --></reference>
  </actor>
  <action><!-- 0..* CodeableConcept Actions controlled by this exception --></action>
  <securityLabel><!-- 0..* Coding Security Labels that define affected resources --></securityLabel>
  <purpose><!-- 0..* Coding Context of activities covered by this exception --></purpose>
  <class><!-- 0..* Coding e.g. Resource Type, Profile, or CDA etc --></class>
  <code><!-- 0..* Coding e.g. LOINC or SNOMED CT code, etc in the content --></code>
  <dataPeriod><!-- 0..1 Period Timeframe for data controlled by this exception --></dataPeriod>
  <data>  <!-- 0..* Data controlled by this exception -->
   <meaning value="[code]"/><!-- 1..1 instance | related | dependents | authoredby -->
   <reference><!-- 1..1 Reference(Any) The actual data reference --></reference>
  </data>
 </except>
</Consent>
```

JSON Template

```
{
  "resourceType" : "Consent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Identifier for this record (external references)
  "status" : "<code>", // R!  draft | proposed | active | rejected | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // Classification of the consent statement - for indexing/retrieval
  "patient" : { Reference(Patient) }, // R!  Who the consent applies to
  "period" : { Period }, // Period that this consent applies
  "dateTime" : "<dateTime>", // When this Consent was created or indexed
  "consentingParty" : [{ Reference(Organization|Patient|Practitioner|
   RelatedPerson) }], // Who is agreeing to the policy and exceptions
  "actor" : [{ // Who|what controlled by this consent (or group, by role)
    "role" : { CodeableConcept }, // R!  How the actor is involved
    "reference" : { Reference(Device|Group|CareTeam|Organization|Patient|
    Practitioner|RelatedPerson) } // R!  Resource for the actor (or group, by role)
  }],
  "action" : [{ CodeableConcept }], // Actions controlled by this consent
  "organization" : [{ Reference(Organization) }], // Custodian of the consent
  // source[x]: Source from which this consent is taken. One of these 3:
  "sourceAttachment" : { Attachment },
  "sourceIdentifier" : { Identifier },
  "sourceReference" : { Reference(Consent|DocumentReference|Contract|
   QuestionnaireResponse) },
  "policy" : [{ // Policies covered by this consent
```

```
    "authority" : "<uri>", // C? Enforcement source for policy
    "uri" : "<uri>" // C? Specific policy covered by this consent
  }],
  "policyRule" : "<uri>", // C? Policy that this consents to
  "securityLabel" : [{ Coding }], // Security Labels that define affected resources
  "purpose" : [{ Coding }], // Context of activities for which the agreement is made
  "dataPeriod" : { Period }, // Timeframe for data controlled by this consent
  "data" : [{ // Data controlled by this consent
    "meaning" : "<code>", // R!  instance | related | dependents | authoredby
    "reference" : { Reference(Any) } // R!  The actual data reference
  }],
  "except" : [{ // Additional rule -  addition or removal of permissions
    "type" : "<code>", // R!  deny | permit
    "period" : { Period }, // Timeframe for this exception
    "actor" : [{ // Who|what controlled by this exception (or group, by role)
      "role" : { CodeableConcept }, // R!  How the actor is involved
      "reference" : { Reference(Device|Group|CareTeam|Organization|Patient|
     Practitioner|RelatedPerson) } // R!  Resource for the actor (or group, by role)
    }],
    "action" : [{ CodeableConcept }], // Actions controlled by this exception
    "securityLabel" : [{ Coding }], // Security Labels that define affected resources
    "purpose" : [{ Coding }], // Context of activities covered by this exception
    "class" : [{ Coding }], // e.g. Resource Type, Profile, or CDA etc
    "code" : [{ Coding }], // e.g. LOINC or SNOMED CT code, etc in the content
    "dataPeriod" : { Period }, // Timeframe for data controlled by this exception
    "data" : [{ // Data controlled by this exception
      "meaning" : "<code>", // R!  instance | related | dependents | authoredby
      "reference" : { Reference(Any) } // R!  The actual data reference
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Consent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Consent.identifier [ Identifier ]; # 0..1 Identifier for this record (external references)
  fhir:Consent.status [ code ]; # 1..1 draft | proposed | active | rejected | inactive | entered-in-error
  fhir:Consent.category [ CodeableConcept ], ... ; # 0..* Classification of the consent statement - for
    indexing/retrieval
  fhir:Consent.patient [ Reference(Patient) ]; # 1..1 Who the consent applies to
  fhir:Consent.period [ Period ]; # 0..1 Period that this consent applies
  fhir:Consent.dateTime [ dateTime ]; # 0..1 When this Consent was created or indexed
  fhir:Consent.consentingParty [ Reference(Organization|Patient|Practitioner|RelatedPerson) ], ... ; # 0..* Who is
    agreeing to the policy and exceptions
  fhir:Consent.actor [ # 0..* Who|what controlled by this consent (or group, by role)
    fhir:Consent.actor.role [ CodeableConcept ]; # 1..1 How the actor is involved
    fhir:Consent.actor.reference [ Reference(Device|Group|CareTeam|Organization|Patient|Practitioner|RelatedPerson) ]; #
     1..1 Resource for the actor (or group, by role)
  ], ...;
  fhir:Consent.action [ CodeableConcept ], ... ; # 0..* Actions controlled by this consent
  fhir:Consent.organization [ Reference(Organization) ], ... ; # 0..* Custodian of the consent
  # Consent.source[x] : 0..1 Source from which this consent is taken. One of these 3
    fhir:Consent.sourceAttachment [ Attachment ]
    fhir:Consent.sourceIdentifier [ Identifier ]
    fhir:Consent.sourceReference [ Reference(Consent|DocumentReference|Contract|QuestionnaireResponse) ]
  fhir:Consent.policy [ # 0..* Policies covered by this consent
    fhir:Consent.policy.authority [ uri ]; # 0..1 Enforcement source for policy
    fhir:Consent.policy.uri [ uri ]; # 0..1 Specific policy covered by this consent
  ], ...;
  fhir:Consent.policyRule [ uri ]; # 0..1 Policy that this consents to
  fhir:Consent.securityLabel [ Coding ], ... ; # 0..* Security Labels that define affected resources
  fhir:Consent.purpose [ Coding ], ... ; # 0..* Context of activities for which the agreement is made
  fhir:Consent.dataPeriod [ Period ]; # 0..1 Timeframe for data controlled by this consent
  fhir:Consent.data [ # 0..* Data controlled by this consent
    fhir:Consent.data.meaning [ code ]; # 1..1 instance | related | dependents | authoredby
    fhir:Consent.data.reference [ Reference(Any) ]; # 1..1 The actual data reference
  ], ...;
  fhir:Consent.except [ # 0..* Additional rule -  addition or removal of permissions
    fhir:Consent.except.type [ code ]; # 1..1 deny | permit
    fhir:Consent.except.period [ Period ]; # 0..1 Timeframe for this exception
    fhir:Consent.except.actor [ # 0..* Who|what controlled by this exception (or group, by role)
      fhir:Consent.except.actor.role [ CodeableConcept ]; # 1..1 How the actor is involved
      fhir:Consent.except.actor.reference [
    Reference(Device|Group|CareTeam|Organization|Patient|Practitioner|RelatedPerson) ]; # 1..1 Resource for the actor
    (or group, by role)
    ], ...;
    fhir:Consent.except.action [ CodeableConcept ], ... ; # 0..* Actions controlled by this exception
    fhir:Consent.except.securityLabel [ Coding ], ... ; # 0..* Security Labels that define affected resources
    fhir:Consent.except.purpose [ Coding ], ... ; # 0..* Context of activities covered by this exception
    fhir:Consent.except.class [ Coding ], ... ; # 0..* e.g. Resource Type, Profile, or CDA etc
    fhir:Consent.except.code [ Coding ], ... ; # 0..* e.g. LOINC or SNOMED CT code, etc in the content
    fhir:Consent.except.dataPeriod [ Period ]; # 0..1 Timeframe for data controlled by this exception
    fhir:Consent.except.data [ # 0..* Data controlled by this exception
```

```
      fhir:Consent.except.data.meaning [ code ]; # 1..1 instance | related | dependents | authoredby
      fhir:Consent.except.data.reference [ Reference(Any) ]; # 1..1 The actual data reference
    ], ...;
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

Consent | I | | DomainResource | A healthcare consumer's policy choices to permits or denies recipients or roles to perform actions for specific purposes and periods of time + Either a Policy or PolicyRule Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..1 | Identifier | Identifier for this record (external references)

status | ?!? | 1..1 | code | draft | proposed | active | rejected | inactive | entered-in-error ConsentState (Required)

category | ? | 0..* | CodeableConcept | Classification of the consent statement - for indexing/retrieval Consent Category Codes (Example)

patient | ? | 1..1 | Reference(Patient) | Who the consent applies to

period | ? | 0..1 | Period | Period that this consent applies

dateTime | ? | 0..1 | dateTime | When this Consent was created or indexed

consentingParty | ? | 0..* | Reference(Organization | Patient | Practitioner | RelatedPerson) | Who is agreeing to the policy and exceptions

actor | ? | 0..* | BackboneElement | Who|what controlled by this consent (or group, by role)

role | | 1..1 | CodeableConcept | How the actor is involved SecurityRoleType (Extensible)

reference | | 1..1 | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson) | Resource for the actor (or group, by role)

action | ? | 0..* | CodeableConcept | Actions controlled by this consent Consent Action Codes (Example)

organization | ? | 0..* | Reference(Organization) | Custodian of the consent

source[x] | ? | 0..1 | | Source from which this consent is taken

sourceAttachment | | | Attachment

sourceIdentifier | | | Identifier

sourceReference | | | Reference(Consent | DocumentReference | Contract | QuestionnaireResponse)

policy | | 0..* | BackboneElement | Policies covered by this consent

authority | I | 0..1 | uri | Enforcement source for policy

uri | I | 0..1 | uri | Specific policy covered by this consent

policyRule | ?I | 0..1 | uri | Policy that this consents to

securityLabel | ? | 0..* | Coding | Security Labels that define affected resources All Security Labels (Extensible)

purpose | ? | 0..* | Coding | Context of activities for which the agreement is made PurposeOfUse (Extensible)

dataPeriod | ? | 0..1 | Period | Timeframe for data controlled by this consent

data | ? | 0..* | BackboneElement | Data controlled by this consent

meaning | ? | 1..1 | code | instance | related | dependents | authoredby ConsentDataMeaning (Required)

reference | ? | 1..1 | Reference(Any) | The actual data reference

except | ? | 0..* | BackboneElement | Additional rule - addition or removal of permissions

type | ? | 1..1 | code | deny | permit ConsentExceptType (Required)

period | ? | 0..1 | Period | Timeframe for this exception

actor | ? | 0..* | BackboneElement | Who|what controlled by this exception (or group, by role)

role | | 1..1 | CodeableConcept | How the actor is involved SecurityRoleType (Extensible)

reference | | 1..1 | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson) | Resource for the actor (or group, by role)

action | ? | 0..* | CodeableConcept | Actions controlled by this exception Consent Action Codes (Example)

securityLabel | ? | 0..* | Coding | Security Labels that define affected resources All Security Labels (Extensible)

purpose | ? | 0..* | Coding | Context of activities covered by this exception PurposeOfUse (Extensible)

class | ? | 0..* | Coding | e.g. Resource Type, Profile, or CDA etc Consent Content Class (Extensible)

code | ? | 0..* | Coding | e.g. LOINC or SNOMED CT code, etc in the content Consent Content Codes (Example)

dataPeriod | ? | 0..1 | Period | Timeframe for data controlled by this exception

data | ? | 0..* | BackboneElement | Data controlled by this exception

meaning | ? | 1..1 | code | instance | related | dependents | authoredby ConsentDataMeaning (Required)

reference | ? | 1..1 | Reference(Any) | The actual data reference

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Consent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Identifier for this record (external references) --></identifier>
 <status value="[code]"/><!-- 1..1 draft | proposed | active | rejected | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept Classification of the consent statement - for indexing/retrieval --></category>
 <patient><!-- 1..1 Reference(Patient) Who the consent applies to --></patient>
 <period><!-- 0..1 Period Period that this consent applies --></period>
 <dateTime value="[dateTime]"/><!-- 0..1 When this Consent was created or indexed -->
 <consentingParty><!-- 0..* Reference(Organization|Patient|Practitioner|
   RelatedPerson) Who is agreeing to the policy and exceptions --></consentingParty>
 <actor>  <!-- 0..* Who|what controlled by this consent (or group, by role) -->
  <role><!-- 1..1 CodeableConcept How the actor is involved --></role>
  <reference><!-- 1..1 Reference(Device|Group|CareTeam|Organization|Patient|
```

```
   Practitioner|RelatedPerson) Resource for the actor (or group, by role) --></reference>
 </actor>
 <action><!-- 0..* CodeableConcept Actions controlled by this consent --></action>
 <organization><!-- 0..* Reference(Organization) Custodian of the consent --></organization>
 <source[x]><!-- 0..1 Attachment|Identifier|Reference(Consent|DocumentReference|
   Contract|QuestionnaireResponse) Source from which this consent is taken --></source[x]>
 <policy>  <!-- 0..* Policies covered by this consent -->
  <authority value="[uri]"/><!--  0..1 Enforcement source for policy -->
  <uri value="[uri]"/><!--  0..1 Specific policy covered by this consent -->
 </policy>
 <policyRule value="[uri]"/><!--  0..1 Policy that this consents to -->
 <securityLabel><!-- 0..* Coding Security Labels that define affected resources --></securityLabel>
 <purpose><!-- 0..* Coding Context of activities for which the agreement is made --></purpose>
 <dataPeriod><!-- 0..1 Period Timeframe for data controlled by this consent --></dataPeriod>
 <data>  <!-- 0..* Data controlled by this consent -->
  <meaning value="[code]"/><!-- 1..1 instance | related | dependents | authoredby -->
  <reference><!-- 1..1 Reference(Any) The actual data reference --></reference>
 </data>
 <except>  <!-- 0..* Additional rule -  addition or removal of permissions -->
  <type value="[code]"/><!-- 1..1 deny | permit -->
  <period><!-- 0..1 Period Timeframe for this exception --></period>
  <actor>  <!-- 0..* Who|what controlled by this exception (or group, by role) -->
   <role><!-- 1..1 CodeableConcept How the actor is involved --></role>
   <reference><!-- 1..1 Reference(Device|Group|CareTeam|Organization|Patient|
     Practitioner|RelatedPerson) Resource for the actor (or group, by role) --></reference>
  </actor>
  <action><!-- 0..* CodeableConcept Actions controlled by this exception --></action>
  <securityLabel><!-- 0..* Coding Security Labels that define affected resources --></securityLabel>
  <purpose><!-- 0..* Coding Context of activities covered by this exception --></purpose>
  <class><!-- 0..* Coding e.g. Resource Type, Profile, or CDA etc --></class>
  <code><!-- 0..* Coding e.g. LOINC or SNOMED CT code, etc in the content --></code>
  <dataPeriod><!-- 0..1 Period Timeframe for data controlled by this exception --></dataPeriod>
  <data>  <!-- 0..* Data controlled by this exception -->
   <meaning value="[code]"/><!-- 1..1 instance | related | dependents | authoredby -->
   <reference><!-- 1..1 Reference(Any) The actual data reference --></reference>
  </data>
 </except>
</Consent>
```

JSON Template

```
{
  "resourceType" : "Consent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Identifier for this record (external references)
  "status" : "<code>", // R!  draft | proposed | active | rejected | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // Classification of the consent statement - for indexing/retrieval
  "patient" : { Reference(Patient) }, // R!  Who the consent applies to
  "period" : { Period }, // Period that this consent applies
  "dateTime" : "<dateTime>", // When this Consent was created or indexed
  "consentingParty" : [{ Reference(Organization|Patient|Practitioner|
   RelatedPerson) }], // Who is agreeing to the policy and exceptions
  "actor" : [{ // Who|what controlled by this consent (or group, by role)
    "role" : { CodeableConcept }, // R!  How the actor is involved
    "reference" : { Reference(Device|Group|CareTeam|Organization|Patient|
    Practitioner|RelatedPerson) } // R!  Resource for the actor (or group, by role)
  }],
  "action" : [{ CodeableConcept }], // Actions controlled by this consent
  "organization" : [{ Reference(Organization) }], // Custodian of the consent
  // source[x]: Source from which this consent is taken. One of these 3:
  "sourceAttachment" : { Attachment },
  "sourceIdentifier" : { Identifier },
  "sourceReference" : { Reference(Consent|DocumentReference|Contract|
   QuestionnaireResponse) },
  "policy" : [{ // Policies covered by this consent
    "authority" : "<uri>", // C? Enforcement source for policy
    "uri" : "<uri>" // C? Specific policy covered by this consent
  }],
  "policyRule" : "<uri>", // C? Policy that this consents to
  "securityLabel" : [{ Coding }], // Security Labels that define affected resources
  "purpose" : [{ Coding }], // Context of activities for which the agreement is made
  "dataPeriod" : { Period }, // Timeframe for data controlled by this consent
  "data" : [{ // Data controlled by this consent
    "meaning" : "<code>", // R!  instance | related | dependents | authoredby
    "reference" : { Reference(Any) } // R!  The actual data reference
  }],
  "except" : [{ // Additional rule -  addition or removal of permissions
    "type" : "<code>", // R!  deny | permit
    "period" : { Period }, // Timeframe for this exception
    "actor" : [{ // Who|what controlled by this exception (or group, by role)
      "role" : { CodeableConcept }, // R!  How the actor is involved
      "reference" : { Reference(Device|Group|CareTeam|Organization|Patient|
     Practitioner|RelatedPerson) } // R!  Resource for the actor (or group, by role)
    }],
    "action" : [{ CodeableConcept }], // Actions controlled by this exception
    "securityLabel" : [{ Coding }], // Security Labels that define affected resources
    "purpose" : [{ Coding }], // Context of activities covered by this exception
    "class" : [{ Coding }], // e.g. Resource Type, Profile, or CDA etc
    "code" : [{ Coding }], // e.g. LOINC or SNOMED CT code, etc in the content
```

```
    "dataPeriod" : { Period }, // Timeframe for data controlled by this exception
    "data" : [{ // Data controlled by this exception
      "meaning" : "<code>", // R!  instance | related | dependents | authoredby
      "reference" : { Reference(Any) } // R!  The actual data reference
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Consent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Consent.identifier [ Identifier ]; # 0..1 Identifier for this record (external references)
  fhir:Consent.status [ code ]; # 1..1 draft | proposed | active | rejected | inactive | entered-in-error
  fhir:Consent.category [ CodeableConcept ], ... ; # 0..* Classification of the consent statement - for
    indexing/retrieval
  fhir:Consent.patient [ Reference(Patient) ]; # 1..1 Who the consent applies to
  fhir:Consent.period [ Period ]; # 0..1 Period that this consent applies
  fhir:Consent.dateTime [ dateTime ]; # 0..1 When this Consent was created or indexed
  fhir:Consent.consentingParty [ Reference(Organization|Patient|Practitioner|RelatedPerson) ], ... ; # 0..* Who is
    agreeing to the policy and exceptions
  fhir:Consent.actor [ # 0..* Who|what controlled by this consent (or group, by role)
    fhir:Consent.actor.role [ CodeableConcept ]; # 1..1 How the actor is involved
    fhir:Consent.actor.reference [ Reference(Device|Group|CareTeam|Organization|Patient|Practitioner|RelatedPerson) ]; #
     1..1 Resource for the actor (or group, by role)
  ], ...;
  fhir:Consent.action [ CodeableConcept ], ... ; # 0..* Actions controlled by this consent
  fhir:Consent.organization [ Reference(Organization) ], ... ; # 0..* Custodian of the consent
  # Consent.source[x] : 0..1 Source from which this consent is taken. One of these 3
    fhir:Consent.sourceAttachment [ Attachment ]
    fhir:Consent.sourceIdentifier [ Identifier ]
    fhir:Consent.sourceReference [ Reference(Consent|DocumentReference|Contract|QuestionnaireResponse) ]
  fhir:Consent.policy [ # 0..* Policies covered by this consent
    fhir:Consent.policy.authority [ uri ]; # 0..1 Enforcement source for policy
    fhir:Consent.policy.uri [ uri ]; # 0..1 Specific policy covered by this consent
  ], ...;
  fhir:Consent.policyRule [ uri ]; # 0..1 Policy that this consents to
  fhir:Consent.securityLabel [ Coding ], ... ; # 0..* Security Labels that define affected resources
  fhir:Consent.purpose [ Coding ], ... ; # 0..* Context of activities for which the agreement is made
  fhir:Consent.dataPeriod [ Period ]; # 0..1 Timeframe for data controlled by this consent
  fhir:Consent.data [ # 0..* Data controlled by this consent
    fhir:Consent.data.meaning [ code ]; # 1..1 instance | related | dependents | authoredby
    fhir:Consent.data.reference [ Reference(Any) ]; # 1..1 The actual data reference
  ], ...;
  fhir:Consent.except [ # 0..* Additional rule -  addition or removal of permissions
    fhir:Consent.except.type [ code ]; # 1..1 deny | permit
    fhir:Consent.except.period [ Period ]; # 0..1 Timeframe for this exception
    fhir:Consent.except.actor [ # 0..* Who|what controlled by this exception (or group, by role)
      fhir:Consent.except.actor.role [ CodeableConcept ]; # 1..1 How the actor is involved
      fhir:Consent.except.actor.reference [
    Reference(Device|Group|CareTeam|Organization|Patient|Practitioner|RelatedPerson) ]; # 1..1 Resource for the actor
    (or group, by role)
    ], ...;
    fhir:Consent.except.action [ CodeableConcept ], ... ; # 0..* Actions controlled by this exception
    fhir:Consent.except.securityLabel [ Coding ], ... ; # 0..* Security Labels that define affected resources
    fhir:Consent.except.purpose [ Coding ], ... ; # 0..* Context of activities covered by this exception
    fhir:Consent.except.class [ Coding ], ... ; # 0..* e.g. Resource Type, Profile, or CDA etc
    fhir:Consent.except.code [ Coding ], ... ; # 0..* e.g. LOINC or SNOMED CT code, etc in the content
    fhir:Consent.except.dataPeriod [ Period ]; # 0..1 Timeframe for data controlled by this exception
    fhir:Consent.except.data [ # 0..* Data controlled by this exception
      fhir:Consent.except.data.meaning [ code ]; # 1..1 instance | related | dependents | authoredby
      fhir:Consent.except.data.reference [ Reference(Any) ]; # 1..1 The actual data reference
    ], ...;
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

6.2.4.1 Terminology Bindings

Path | Definition | Type | Reference

Consent.status | Indicates the state of the consent | Required | ConsentState

Consent.category | A classification of the type of consents found in a consent statement | Example | Consent Category Codes

Consent.actor.role Consent.except.actor.role | How an actor is involved in the consent considerations | Extensible | SecurityRoleType

Consent.action Consent.except.action | Detailed codes for the consent action. | Example | Consent Action Codes

Consent.securityLabel Consent.except.securityLabel | Security Labels from the Healthcare Privacy and Security Classification System. | Extensible | All Security Labels

Consent.purpose Consent.except.purpose | What purposes of use are controlled by this exception. If more than one label is specified, operations must have all the specified labels | Extensible | PurposeOfUse
Consent.data.meaning Consent.except.data.meaning | How a resource reference is interpreted when testing consent restrictions | Required | ConsentDataMeaning
Consent.except.type | How an exception statement is applied, such as adding additional consent or removing consent | Required | ConsentExceptType
Consent.except.class | The class (type) of information a consent rule covers | Extensible | Consent Content Class
Consent.except.code | If this code is found in an instance, then the exception applies | Example | Consent Content Codes

6.2.4.2 Constraints

* ppc-1: Either a Policy or PolicyRule (expression : policy.exists() or policyRule.exists())

6.2.5 Policies

This specification defines 2 magic values for consent policyRule:

URI | Description
http://hl7.org/fhir/ConsentPolicy/opt-out | The basic 'deny' policy is to grant no authority for data access or use. No actions are approved, unless they are explicitly detailed in the exceptions
http://hl7.org/fhir/ConsentPolicy/opt-in | The basic 'permit' policy is to grant generally acceptable authority for data access and use. All generally acceptable actions are approved, unless they are explicitly detailed in the exceptions

Other jurisdictions (e.g. HL7 Affiliates) may define their additional lists of consent policy URI values that represent consent policies established by law or regulation

The Consent resource has a reference to a single policyRule. Many organizations will work in a context where multiple different consent regulations and policies apply. In these cases, the single policy rule reference refers to a policy document that resolves and reconciles the various policies, and presents a single policy for patient consent. If it is still necessary to track which of the underlying policies an exception is make in regard to, the policy may be used.

6.2.6 General Model

The following is the general model of Privacy Consent Directives.

There are context setting parameters:

* Who - The patient

* What - The data - specific resources are listed, empty list means all data covered by the consent.

* Where - The domain and authority - what is the location boundary and authority boundary of this consent

* When - The issued or captured

* When - The timeframe for which the Consent applies

* How - The actions covered. (such as purposes of use that are covered)

* Whom - The recipient are grantees by the consent.

A Privacy Consent may transition through many states including: that no consent has been sought, consent has been proposed, consent has been rejected, and consent approved.

There are set of patterns.

* No consent: All settings need a policy for when no consent has been captured. Often this allows treatment only.;

* Opt-out: No sharing allowed for the specified domain, location, actions, and purposes;

* Opt-out with exceptions: No sharing allowed, with some exceptions where it is allowed. Example: Withhold Authorization for Treatment except for Emergency Treatment;

* Opt-in: Sharing for some purpose of use is authorized Sharing allowed for Treatment, Payment, and normal Operations; and

* Opt-in with restrictions: Sharing allowed, but the patient may make exceptions (See the Canadian examples).

For each of these patterns (positive or negative pattern), there can be exceptions. These exceptions are explicitly recorded in the

except

6.2.7 Realm specifics

6.2.7.1 US Realm sample Use-Cases

Five categories of Privacy Consent Directives are described in the Office of the National Coordinator for Health Information (ONC) Consent Directives Document released March 31, 2010, and include the following US-specific “Core consent options” for electronic exchange:

* No consent: Health information of patients is automatically included—patients cannot opt out;

* Opt-out: Default is for health information of patients to be included automatically, but the patient can opt out completely;

* Opt-out with exceptions: Default is for health information of patients to be included, but the patient can opt out completely or allow only select data to be included;

* Opt-in: Default is that no patient health information is included; patients must actively express consent to be included, but if they do so then their information must be all in or all out; and

* Opt-in with restrictions: Default is that no patient health information is made available, but the patient may allow a subset of select data to be included.

A common exception is to explicitly exclude or explicitly include a period of time.

6.2.7.2 Canada Realm sample Use-Cases

The following scenarios are based on existing jurisdictional policy and are realized in existing systems in Canada. The default policy is one of implied consent for the provision of care, so these scenarios all deal with withdrawal or withholding consent for that purpose. In other jurisdictions, where an express consent model is used (Opt-In), these examples would contain the phrase "consent to" rather than "withhold" or "withdraw" consent for.

* Withhold or withdraw consent for disclosure of records related to specific domain (e.g. DI, LAB, etc.)

* Withhold or withdraw consent for disclosure of a specific record (e.g. Lab Order/Result)

* Withhold or withdraw consent for disclosure to a specific provider organization

* Withhold or withdraw consent for disclosure to a specific provider agent (an individual within an organization)

* Withhold or withdraw consent for disclosure of records that were authored by a specific organization (or service delivery location).

* Combinations of the above

6.2.7.3 Non Treatment Use-Cases

Also shown is an example where a Patient has authorized disclosure to a specific individual for purposes directed by the patient (possibly not a treatment case).

6.2.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
action | token | Actions controlled by this consent | Consent.action | Consent.except.action
actor | reference | Resource for the actor (or group, by role) | Consent.actor.reference | Consent.except.actor.reference (Practitioner, Group, Organization, CareTeam, Device, Patient, RelatedPerson)
category | token | Classification of the consent statement - for indexing/retrieval | Consent.category
consentor | reference | Who is agreeing to the policy and exceptions | Consent.consentingParty (Practitioner, Organization, Patient, RelatedPerson)
data | reference | The actual data reference | Consent.data.reference | Consent.except.data.reference (Any)
date | date | When this Consent was created or indexed | Consent.dateTime | 18 Resources
identifier | token | Identifier for this record (external references) | Consent.identifier | 26 Resources
organization | reference | Custodian of the consent | Consent.organization (Organization)
patient | reference | Who the consent applies to | Consent.patient (Patient) | 31 Resources
period | date | Period that this consent applies | Consent.period
purpose | token | Context of activities for which the agreement is made | Consent.purpose | Consent.except.purpose
securitylabel | token | Security Labels that define affected resources | Consent.securityLabel | Consent.except.securityLabel
source | reference | Source from which this consent is taken | Consent.source (Consent, Contract, QuestionnaireResponse, DocumentReference)
status | token | draft | proposed | active | rejected | inactive | entered-in-error | Consent.status

6.2.9 Resource Consent - Examples

URL: https://hl7.org/fhir/STU3/consent-examples.html

These are various examples that have been used to drive the creaion of the Consent resource.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.2.9 Resource Consent - Examples

Community Based Collaborative Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Example List:

General Consent Example | consent-example-basic | XML | JSON | Turtle
Consent withholding access | consent-example-Out | XML | JSON | Turtle
Consent withholding access to data for Treatment exept for Emergency Treatment | consent-example-Emergency | XML | JSON | Turtle
Withhold or withdraw consent for disclosure of records related to specific domain (e.g. DI, LAB, etc.) | consent-example-notThis | XML | JSON | Turtle
Withhold or withdraw consent for disclosure of a specific timeframe | consent-example-notTime | XML | JSON | Turtle
Withhold or withdraw consent for disclosure to a specific provider organization | consent-example-notOrg | XML | JSON | Turtle
Withhold or withdraw consent for disclosure to a specific provider agent (an individual within an organization) | consent-example-notThem | XML | JSON | Turtle
Withhold or withdraw consent for disclosure of records that were authored by a specific organization (or service delivery location). | consent-example-notAuthor | XML | JSON | Turtle
Patient grants access to a specified individual for read-only access | consent-example-grantor | XML | JSON | Turtle
Example of Patients Know Best Usage | consent-example-pkb | XML | JSON | Turtle
Template for recording a Smart on FHIR Authorization | consent-example-smartonfhir | XML | JSON | Turtle
Consent Example with a signature | consent-example-signature | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

6.2.10 Resource Consent - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/consent-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.2.10 Resource Consent - Detailed Descriptions

Community Based Collaborative Care Work Group | Maturity Level: 1 | Trial Use | Compartments: Patient

Detailed Descriptions for the elements in the Consent resource.

Consent
Definition | A record of a healthcare consumer's policy choices, which permits or denies identified recipient(s) or recipient role(s) to perform one or more actions within a given policy context, for specific purposes and periods of time.
Control | 1..1
Comments | Broadly, there are 3 key areas of consent for patients: Consent around sharing information (aka Privacy Consent Directive - Authorization to Collect, Use, or Disclose information), consent for specific treatment, or kinds of treatment, and general advance care directives.
Invariants | Defined on this element ppc-1: Either a Policy or PolicyRule (expression : policy.exists() or policyRule.exists(), xpath: exists(f:policy) or exists(f:policyRule))
Consent.identifier
Definition | Unique identifier for this copy of the Consent Statement.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
Comments | This identifier is identifies this copy of the consent. Where this identifier is also used elsewhere as the identifier for a consent record (e.g. a CDA consent document) then the consent details are expected to be the same.
Consent.status
Definition | Indicates the current state of this consent.
Control | 1..1
Terminology Binding | ConsentState (Required)
Type | code
Is Modifier | true
Requirements | The Consent Directive that is pointed to might be in various lifecycle states, e.g., a revoked Consent Directive.
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes rejected and entered-in-error that mark the Consent as not currently valid.
Consent.category
Definition | A classification of the type of consents found in the statement. This element supports indexing and retrieval of consent statements.
Control | 0..*
Terminology Binding | Consent Category Codes (Example)
Type | CodeableConcept
Summary | true
Consent.patient
Definition | The patient/healthcare consumer to whom this consent applies.
Control | 1..1
Type | Reference(Patient)
Summary | true
Comments | Commonly, the patient the consent pertains to is the author, but for young and old people, it may be some other person.
Consent.period
Definition | Relevant time or time-period when this Consent is applicable.
Control | 0..1
Type | Period
Meaning if Missing | all date ranges included
Summary | true
Comments | This can be a subset of the period of the original statement.
Consent.dateTime
Definition | When this Consent was issued / created / indexed.
Control | 0..1
Type | dateTime
Summary | true
Comments | This is not the time of the original consent, but the time that this statement was made or derived.
Consent.consentingParty
Definition | Either the Grantor, which is the entity responsible for granting the rights listed in a Consent Directive or the Grantee, which is the entity responsible for complying with the Consent Directive, including any obligations or limitations on authorizations and enforcement of prohibitions.
Control | 0..*
Type | Reference(Organization | Patient | Practitioner | RelatedPerson)
Alternate Names | consentor
Summary | true

Comments | Commonly, the patient the consent pertains to is the consentor, but particularly for young and old people, it may be some other person - e.g. a legal guardian.
Consent.actor
Definition | Who or what is controlled by this consent. Use group to identify a set of actors by some property they share (e.g. 'admitting officers').
Control | 0..*
Meaning if Missing | There is no specific actor associated with the consent
Summary | true
Consent.actor.role
Definition | How the individual is involved in the resources content that is described in the consent.
Control | 1..1
Terminology Binding | SecurityRoleType (Extensible)
Type | CodeableConcept
Consent.actor.reference
Definition | The resource that identifies the actor. To identify a actors by type, use group to identify a set of actors by some property they share (e.g. 'admitting officers').
Control | 1..1
Type | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson)
Consent.action
Definition | Actions controlled by this consent.
Control | 0..*
Terminology Binding | Consent Action Codes (Example)
Type | CodeableConcept
Meaning if Missing | all actions
Summary | true
Comments | Note that this is the direct action (not the grounds for the action covered in the purpose element). At present, the only action in the understood and tested scope of this resource is 'read'.
Consent.organization
Definition | The organization that manages the consent, and the framework within which it is executed.
Control | 0..*
Type | Reference(Organization)
Alternate Names | custodian
Summary | true
Consent.source[x]
Definition | The source on which this consent statement is based. The source might be a scanned original paper form, or a reference to a consent that links back to such a source, a reference to a document repository (e.g. XDS) that stores the original consent document.
Control | 0..1
Type | Attachment|Identifier|Reference(Consent | DocumentReference | Contract | QuestionnaireResponse)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | The source can be contained inline (Attachment), referenced directly (Consent), referenced in a consent repository (DocumentReference), or simply by an identifier (Identifier), e.g. a CDA document id.
Consent.policy
Definition | The references to the policies that are included in this consent scope. Policies may be organizational, but are often defined jurisdictionally, or in law.
Control | 0..*
Consent.policy.authority
Definition | Entity or Organization having regulatory jurisdiction or accountability for enforcing policies pertaining to Consent Directives.
Control | 0..1
Type | uri
Invariants | Affect this element ppc-1: Either a Policy or PolicyRule (expression : policy.exists() or policyRule.exists(), xpath: exists(f:policy) or exists(f:policyRule))
Consent.policy.uri
Definition | The references to the policies that are included in this consent scope. Policies may be organizational, but are often defined jurisdictionally, or in law.
Control | 0..1
Type | uri
Comments | This element is for discoverability / documentation, and does not modify or qualify the policy rules.
Invariants | Affect this element ppc-1: Either a Policy or PolicyRule (expression : policy.exists() or policyRule.exists(), xpath: exists(f:policy) or exists(f:policyRule))
Consent.policyRule
Definition | A referece to the specific computable policy.
Control | 0..1
Type | uri
Requirements | Might be a unique identifier of a policy set in XACML, or other rules engine.
Summary | true
Comments | If the policy reference is not known, the resource cannot be processed. Where the reference is absent, there is no particular policy other than what is expressed directly in the consent resource.
Invariants | Affect this element ppc-1: Either a Policy or PolicyRule (expression : policy.exists() or policyRule.exists(), xpath: exists(f:policy) or exists(f:policyRule))
Consent.securityLabel
Definition | A set of security labels that define which resources are controlled by this consent. If more than one label is specified, all resources must have all the specified labels.
Control | 0..*

Terminology Binding | All Security Labels (Extensible)
Type | Coding
Summary | true
Comments | If the consent specifies a security label of "R" then it applies to all resources that are labeled "R" or lower. E.g. for Confidentiality, it's a high water mark. For other kinds of security labels, subsumption logic applies. Not all of the security labels make sense for use in this element (may define a narrower value set?).
Consent.purpose
Definition | The context of the activities a user is taking - why the user is accessing the data - that are controlled by this consent.
Control | 0..*
Terminology Binding | PurposeOfUse (Extensible)
Type | Coding
Summary | true
Comments | This element is for discoverability / documentation, and does not modify or qualify the policy (e.g. the policy itself describes the purposes for which it applies).
Consent.dataPeriod
Definition | Clinical or Operational Relevant period of time that bounds the data controlled by this consent.
Control | 0..1
Type | Period
Summary | true
Comments | This has a different sense to the Consent.period - that is when the consent agreement holds. This is the time period of the data that is controlled by the agreement.
Consent.data
Definition | The resources controlled by this consent, if specific resources are referenced.
Control | 0..*
Meaning if Missing | all data
Summary | true
Consent.data.meaning
Definition | How the resource reference is interpreted when testing consent restrictions.
Control | 1..1
Terminology Binding | ConsentDataMeaning (Required)
Type | code
Summary | true
Consent.data.reference
Definition | A reference to a specific resource that defines which resources are covered by this consent.
Control | 1..1
Type | Reference(Any)
Summary | true
Consent.except
Definition | An exception to the base policy of this consent. An exception can be an addition or removal of access permissions.
Control | 0..*
Summary | true
Consent.except.type
Definition | Action to take - permit or deny - when the exception conditions are met.
Control | 1..1
Terminology Binding | ConsentExceptType (Required)
Type | code
Summary | true
Consent.except.period
Definition | The timeframe in this exception is valid.
Control | 0..1
Type | Period
Summary | true
Consent.except.actor
Definition | Who or what is controlled by this Exception. Use group to identify a set of actors by some property they share (e.g. 'admitting officers').
Control | 0..*
Meaning if Missing | There is no specific actor associated with the exception
Summary | true
Consent.except.actor.role
Definition | How the individual is involved in the resources content that is described in the exception.
Control | 1..1
Terminology Binding | SecurityRoleType (Extensible)
Type | CodeableConcept
Consent.except.actor.reference
Definition | The resource that identifies the actor. To identify a actors by type, use group to identify a set of actors by some property they share (e.g. 'admitting officers').
Control | 1..1
Type | Reference(Device | Group | CareTeam | Organization | Patient | Practitioner | RelatedPerson)
Consent.except.action
Definition | Actions controlled by this Exception.
Control | 0..*
Terminology Binding | Consent Action Codes (Example)
Type | CodeableConcept
Meaning if Missing | all actions

Summary | true
Comments | Note that this is the direct action (not the grounds for the action covered in the purpose element). At present, the only action in the understood and tested scope of this resource is 'read'.
Consent.except.securityLabel
Definition | A set of security labels that define which resources are controlled by this exception. If more than one label is specified, all resources must have all the specified labels.
Control | 0..*
Terminology Binding | All Security Labels (Extensible)
Type | Coding
Summary | true
Comments | If the consent specifies a security label of "R" then it applies to all resources that are labeled "R" or lower. E.g. for Confidentiality, it's a high water mark. For other kinds of security labels, subsumption logic applies. Not all of the security labels make sense for use in this element (may define a narrower value set?).
Consent.except.purpose
Definition | The context of the activities a user is taking - why the user is accessing the data - that are controlled by this exception.
Control | 0..*
Terminology Binding | PurposeOfUse (Extensible)
Type | Coding
Summary | true
Comments | E.g. if the purpose is 'research', then the operational context must be research, in order for the consent to apply. Not all of the security labels make sense for use in this element (may define a narrower value set?).
Consent.except.class
Definition | The class of information covered by this exception. The type can be a FHIR resource type, a profile on a type, or a CDA document, or some other type that indicates what sort of information the consent relates to.
Control | 0..*
Terminology Binding | Consent Content Class (Extensible)
Type | Coding
Summary | true
Comments | Multiple types are or'ed together. The intention of the contentType element is that the codes refer to profiles or document types defined in a standard or an implementation guide somewhere.
Consent.except.code
Definition | If this code is found in an instance, then the exception applies.
Control | 0..*
Terminology Binding | Consent Content Codes (Example)
Type | Coding
Summary | true
Comments | Typical use of this is a Document code with class = CDA.
Consent.except.dataPeriod
Definition | Clinical or Operational Relevant period of time that bounds the data controlled by this exception.
Control | 0..1
Type | Period
Summary | true
Comments | This has a different sense to the Consent.period - that is when the consent agreement holds. This is the time period of the data that is controlled by the agreement.
Consent.except.data
Definition | The resources controlled by this exception, if specific resources are referenced.
Control | 0..*
Meaning if Missing | all data
Summary | true
Consent.except.data.meaning
Definition | How the resource reference is interpreted when testing consent restrictions.
Control | 1..1
Terminology Binding | ConsentDataMeaning (Required)
Type | code
Summary | true
Consent.except.data.reference
Definition | A reference to a specific resource that defines which resources are covered by this consent.
Control | 1..1
Type | Reference(Any)
Summary | true

6.2.11 Resource Consent - Mappings

URL: https://hl7.org/fhir/STU3/consent-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.2.11 Resource Consent - Mappings

Community Based Collaborative Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the consent resource.

6.2.11.1 Mappings for RIM Mapping (http://hl7.org/v3)

Consent | FinancialConsent
identifier | .id
status | .statusCode
category | CNTRCT
patient | Role
period | FinancialConsent activityTime
dateTime | FinancialConsent effectiveTime
consentingParty
actor
role
reference
action
organization
source[x]
policy
authority
uri
policyRule
securityLabel
purpose
dataPeriod
data | Role
meaning
reference
except
type
period
actor
role
reference
action
securityLabel
purpose
class
code
dataPeriod
data | Role
meaning
reference

6.2.11.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Consent
identifier
status | HL7 Table 0498 - Consent Status
category
patient
period
dateTime
consentingParty
actor
role
reference
action
organization
source[x]
policy
authority
uri
policyRule
securityLabel
purpose
dataPeriod
data
meaning
reference
except
type

period
actor
role
reference
action
securityLabel
purpose
class
code
dataPeriod
data
meaning
reference

6.2.12 Resource Consent - Design Notes

URL: https://hl7.org/fhir/STU3/consent-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.2.12 Resource Consent - Design Notes

to do

6.2.13 Resource Consent - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/consent-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.2.13 Resource Consent - Extensions & Profiles

Community Based Collaborative Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Consent resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

6.2.13.1 Profiles

No Profiles defined for this resource

6.2.13.2 Extensions

consent-NotificationEndpoint | NotificationEndpoint | for Consent HL7 Extensions
consent-Witness | Witness | for Consent HL7 Extensions
consent-location | location | for Consent HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

6.2.13.3 Search Extensions

No Search Extensions defined for this resource

6.2.14 Consent HL7 Extensions

URL: https://hl7.org/fhir/STU3/consent-extensions.html

Defines common extensions used with or related to the Consent resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.2.14 Consent HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

6.2.14.1 Content

Extensions:
consent-NotificationEndpoint | NotificationEndpoint : Endpoint for sending Disclosure notifications in the form of FHIR AuditEvent records.
consent-Witness | Witness : Any witness to the consent.
consent-location | location : Restricts this exception to only apply a specific location as defined.

6.3 Resource Provenance - Content

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Security Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Provenance of a resource is a record that describes entities and processes involved in producing and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing authenticity, enabling trust, and allowing reproducibility. Provenance assertions are a form of contextual metadata and can themselves become important records with their own provenance. Provenance statement indicates clinical significance in terms of confidence in authenticity, reliability, and trustworthiness, integrity, and stage in lifecycle (e.g. Document Completion - has the artifact been legally authenticated), all of which may impact security, privacy, and trust policies.

6.3.1 Scope and Usage

The Provenance resource tracks information about the activity that created a version of a resource, including the entities and agents involved in producing a resource. This information can be used to form assessments about its quality, reliability or trustworthiness, or to provide pointers for where to go to further investigate the origins of the resource and the information in it.

Provenance resources are a record-keeping assertion that gathers information about the context in which the information in a resource was obtained. Provenance resources are prepared by the application that initiates the create/update etc. of the resource. An AuditEvent resource contains overlapping information, but is created as events occur, to track and audit the events. AuditEvent resources are often (though not exclusively) created by the application responding to the read/query/create/update/etc. event.

6.3.2 Boundaries and Relationships

Many other FHIR resources contain some elements that represent information about how the resource was obtained, and therefore they overlap with the functionality of the Provenance resource. These properties in other resources should always be used in preference to the Provenance resource, and the Provenance resource should be used where additional information is required, though overlap can occur.

The relationship between a resource and its provenance is established by a reference from the provenance resource to its target. In this way, provenance may be provided about any resource or version, including past versions. There may be multiple provenance records for a given resource or version of a resource.

6.3.3 Background and Context

The Provenance resource is based on the W3C Provenance specification , and mappings are provided. The Provenance resource is tailored to fit the FHIR use-cases for provenance more directly. In terms of W3C Provenance the FHIR Provenance resources covers "Generation" of "Entity" with respect to FHIR defined resources for creation or updating; whereas AuditEvent covers "Usage" of "Entity" and all other "Activity" as defined in W3C Provenance.

The W3C Provenance Specification has the following fundamental model:

Where:

* Target - An entity that is a FHIR resource instance that is created, updated or deleted.

* Entity - An entity is a physical, digital, conceptual or other kind of thing with some fixed aspects; entities may be real or imaginary.

* Agent - An agent is something that bears some form of responsibility for an activity taking place, for the existence of an entity, or for another agent's activity.

* Activity - An activity is something that occurs over a time period and acts upon or with entities. It may include consuming, processing, transforming, modifying, relocating, using, or generating entities.

The Provenance resource corresponds to a single activity that identifies a set of resources (target) generated by the activity. The activity also references other entities (entity) that were used and the agents (agent) that were associated with the activity. To record multiple activities that resulted in one (target), record each (activity) in independent Provenance records all pointing at that (target).

The Provenance resource depends upon having References to all the resources, entities, and agents involved in the activity. These References need not be resolvable. The references must provide a unique and ambiguous identification. If a resource, entity, or agent can have different versions that must be identified, then the Reference must have versioning information included.

Versioning and unique identification are not mandated for all systems that provide Resources, entities, and agents. But, inclusion of Provenance requirements may introduce requirements for versioning and unique identification on those systems

The Provenance resource is based on leveraging the W3C Provenance specification to represent HL7 support of provenance throughout its standards and explicitly modelled as functional capabilities in ISO/HL7 10781 EHR System Functional Model Release 2 and ISO 21089 Trusted End-to-End Information Flows. Mappings are provided. The Provenance resource is tailored to fit the FHIR use-cases for provenance more directly. In terms of W3C Provenance the FHIR Provenance resources covers "Generation" of "Entity" with respect to FHIR defined resources for creation or updating; whereas AuditEvent covers "Usage" of "Entity" and all other "Activity" as defined in W3C Provenance.

This resource is referenced by DeviceRequest, MedicationAdministration, MedicationDispense, MedicationRequest, ProcedureRequest, ReferralRequest and Task

6.3.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Provenance | | | DomainResource | Who, What, When for a set of resources Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
target | ? | 1..* | Reference(Any) | Target Reference(s) (usually version specific)
period | | 0..1 | Period | When the activity occurred
recorded | ? | 1..1 | instant | When the activity was recorded / updated
policy | | 0..* | uri | Policy or plan the activity was defined by
location | | 0..1 | Reference(Location) | Where the activity occurred, if relevant
reason | | 0..* | Coding | Reason the activity is occurring PurposeOfUse (Extensible)
activity | | 0..1 | Coding | Activity that occurred ProvenanceActivityType (Extensible)
agent | | 1..* | BackboneElement | Actor involved
role | ? | 0..* | CodeableConcept | What the agents role was SecurityRoleType (Extensible)
who[x] | ? | 1..1 | | Who participated
whoUri | | | uri
whoReference | | | Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
onBehalfOf[x] | | 0..1 | | Who the agent is representing
onBehalfOfUri | | | uri
onBehalfOfReference | | | Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
relatedAgentType | | 0..1 | CodeableConcept | Type of relationship between agents v3 Code System RoleLinkType (Example)
entity | | 0..* | BackboneElement | An entity used in this activity
role | ? | 1..1 | code | derivation | revision | quotation | source | removal ProvenanceEntityRole (Required)
what[x] | ? | 1..1 | | Identity of entity
whatUri | | | uri
whatReference | | | Reference(Any)
whatIdentifier | | | Identifier
agent | | 0..* | see agent | Entity is attributed to this agent
signature | | 0..* | Signature | Signature on target
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Provenance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <target><!-- 1..* Reference(Any) Target Reference(s) (usually version specific) --></target>
 <period><!-- 0..1 Period When the activity occurred --></period>
 <recorded value="[instant]"/><!-- 1..1 When the activity was recorded / updated -->
 <policy value="[uri]"/><!-- 0..* Policy or plan the activity was defined by -->
 <location><!-- 0..1 Reference(Location) Where the activity occurred, if relevant --></location>
 <reason><!-- 0..* Coding Reason the activity is occurring --></reason>
 <activity><!-- 0..1 Coding Activity that occurred --></activity>
 <agent>  <!-- 1..* Actor involved -->
  <role><!-- 0..* CodeableConcept What the agents role was --></role>
  <who[x]><!-- 1..1 uri|Reference(Practitioner|RelatedPerson|Patient|Device|
    Organization) Who participated --></who[x]>
  <onBehalfOf[x]><!-- 0..1 uri|Reference(Practitioner|RelatedPerson|Patient|
    Device|Organization) Who the agent is representing --></onBehalfOf[x]>
  <relatedAgentType><!-- 0..1 CodeableConcept Type of relationship between agents --></relatedAgentType>
 </agent>
 <entity>  <!-- 0..* An entity used in this activity -->
  <role value="[code]"/><!-- 1..1 derivation | revision | quotation | source | removal -->
  <what[x]><!-- 1..1 uri|Reference(Any)|Identifier Identity of entity --></what[x]>
  <agent><!-- 0..* Content as for Provenance.agent Entity is attributed to this agent --></agent>
 </entity>
 <signature><!-- 0..* Signature Signature on target --></signature>
</Provenance>
```

JSON Template

```
{
  "resourceType" : "Provenance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
```

```
  "target" : [{ Reference(Any) }], // R!  Target Reference(s) (usually version specific)
  "period" : { Period }, // When the activity occurred
  "recorded" : "<instant>", // R!  When the activity was recorded / updated
  "policy" : ["<uri>"], // Policy or plan the activity was defined by
  "location" : { Reference(Location) }, // Where the activity occurred, if relevant
  "reason" : [{ Coding }], // Reason the activity is occurring
  "activity" : { Coding }, // Activity that occurred
  "agent" : [{ // R!  Actor involved
    "role" : [{ CodeableConcept }], // What the agents role was
    // who[x]: Who participated. One of these 2:
    "whoUri" : "<uri>",
    "whoReference" : { Reference(Practitioner|RelatedPerson|Patient|Device|
    Organization) },
    // onBehalfOf[x]: Who the agent is representing. One of these 2:
    "onBehalfOfUri" : "<uri>",
    "onBehalfOfReference" : { Reference(Practitioner|RelatedPerson|Patient|Device|
    Organization) },
    "relatedAgentType" : { CodeableConcept } // Type of relationship between agents
  }],
  "entity" : [{ // An entity used in this activity
    "role" : "<code>", // R!  derivation | revision | quotation | source | removal
    // what[x]: Identity of entity. One of these 3:
    "whatUri" : "<uri>",
    "whatReference" : { Reference(Any) },
    "whatIdentifier" : { Identifier },
    "agent" : [{ Content as for Provenance.agent }] // Entity is attributed to this agent
  }],
  "signature" : [{ Signature }] // Signature on target
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Provenance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Provenance.target [ Reference(Any) ], ... ; # 1..* Target Reference(s) (usually version specific)
  fhir:Provenance.period [ Period ]; # 0..1 When the activity occurred
  fhir:Provenance.recorded [ instant ]; # 1..1 When the activity was recorded / updated
  fhir:Provenance.policy [ uri ], ... ; # 0..* Policy or plan the activity was defined by
  fhir:Provenance.location [ Reference(Location) ]; # 0..1 Where the activity occurred, if relevant
  fhir:Provenance.reason [ Coding ], ... ; # 0..* Reason the activity is occurring
  fhir:Provenance.activity [ Coding ]; # 0..1 Activity that occurred
  fhir:Provenance.agent [ # 1..* Actor involved
    fhir:Provenance.agent.role [ CodeableConcept ], ... ; # 0..* What the agents role was
    # Provenance.agent.who[x] : 1..1 Who participated. One of these 2
      fhir:Provenance.agent.whoUri [ uri ]
      fhir:Provenance.agent.whoReference [ Reference(Practitioner|RelatedPerson|Patient|Device|Organization) ]
    # Provenance.agent.onBehalfOf[x] : 0..1 Who the agent is representing. One of these 2
      fhir:Provenance.agent.onBehalfOfUri [ uri ]
      fhir:Provenance.agent.onBehalfOfReference [ Reference(Practitioner|RelatedPerson|Patient|Device|Organization) ]
    fhir:Provenance.agent.relatedAgentType [ CodeableConcept ]; # 0..1 Type of relationship between agents
  ], ...;
  fhir:Provenance.entity [ # 0..* An entity used in this activity
    fhir:Provenance.entity.role [ code ]; # 1..1 derivation | revision | quotation | source | removal
    # Provenance.entity.what[x] : 1..1 Identity of entity. One of these 3
      fhir:Provenance.entity.whatUri [ uri ]
      fhir:Provenance.entity.whatReference [ Reference(Any) ]
      fhir:Provenance.entity.whatIdentifier [ Identifier ]
    fhir:Provenance.entity.agent [ See Provenance.agent ], ... ; # 0..* Entity is attributed to this agent
  ], ...;
  fhir:Provenance.signature [ Signature ], ... ; # 0..* Signature on target
]
```

Changes since DSTU2

Provenance

Provenance.reason | Type changed from CodeableConcept to Coding

Provenance.activity | Type changed from CodeableConcept to Coding Change value set from http://hl7.org/fhir/ValueSet/v3-ProvenanceEventCurrentState to http://hl7.org/fhir/ValueSet/provenance-activity-type

Provenance.agent | Min Cardinality changed from 0 to 1

Provenance.agent.role | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to * Type changed from Coding to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/provenance-agent-role to http://hl7.org/fhir/ValueSet/security-role-type

Provenance.agent.who[x] | Added Element

Provenance.agent.onBehalfOf[x] | Added Element

Provenance.agent.relatedAgentType | Added Element

Provenance.entity.what[x] | Added Element

Provenance.entity.agent | Max Cardinality changed from 1 to *

Provenance.agent.actor | deleted

Provenance.agent.userId | deleted

Provenance.agent.relatedAgent | deleted

Provenance.entity.type | deleted

Provenance.entity.reference | deleted

Provenance.entity.display | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (1 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
Provenance | | | DomainResource | Who, What, When for a set of resources Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
target | ? | 1..* | Reference(Any) | Target Reference(s) (usually version specific)
period | | 0..1 | Period | When the activity occurred
recorded | ? | 1..1 | instant | When the activity was recorded / updated
policy | | 0..* | uri | Policy or plan the activity was defined by
location | | 0..1 | Reference(Location) | Where the activity occurred, if relevant
reason | | 0..* | Coding | Reason the activity is occurring PurposeOfUse (Extensible)
activity | | 0..1 | Coding | Activity that occurred ProvenanceActivityType (Extensible)
agent | | 1..* | BackboneElement | Actor involved
role | ? | 0..* | CodeableConcept | What the agents role was SecurityRoleType (Extensible)
who[x] | ? | 1..1 | | Who participated
whoUri | | | uri
whoReference | | | Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
onBehalfOf[x] | | 0..1 | | Who the agent is representing
onBehalfOfUri | | | uri
onBehalfOfReference | | | Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
relatedAgentType | | 0..1 | CodeableConcept | Type of relationship between agents v3 Code System RoleLinkType (Example)
entity | | 0..* | BackboneElement | An entity used in this activity
role | ? | 1..1 | code | derivation | revision | quotation | source | removal ProvenanceEntityRole (Required)
what[x] | ? | 1..1 | | Identity of entity
whatUri | | | uri
whatReference | | | Reference(Any)
whatIdentifier | | | Identifier
agent | | 0..* | see agent | Entity is attributed to this agent
signature | | 0..* | Signature | Signature on target
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Provenance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <target><!-- 1..* Reference(Any) Target Reference(s) (usually version specific) --></target>
 <period><!-- 0..1 Period When the activity occurred --></period>
 <recorded value="[instant]"/><!-- 1..1 When the activity was recorded / updated -->
 <policy value="[uri]"/><!-- 0..* Policy or plan the activity was defined by -->
 <location><!-- 0..1 Reference(Location) Where the activity occurred, if relevant --></location>
 <reason><!-- 0..* Coding Reason the activity is occurring --></reason>
 <activity><!-- 0..1 Coding Activity that occurred --></activity>
 <agent>  <!-- 1..* Actor involved -->
  <role><!-- 0..* CodeableConcept What the agents role was --></role>
  <who[x]><!-- 1..1 uri|Reference(Practitioner|RelatedPerson|Patient|Device|
    Organization) Who participated --></who[x]>
  <onBehalfOf[x]><!-- 0..1 uri|Reference(Practitioner|RelatedPerson|Patient|
    Device|Organization) Who the agent is representing --></onBehalfOf[x]>
  <relatedAgentType><!-- 0..1 CodeableConcept Type of relationship between agents --></relatedAgentType>
 </agent>
 <entity>  <!-- 0..* An entity used in this activity -->
  <role value="[code]"/><!-- 1..1 derivation | revision | quotation | source | removal -->
  <what[x]><!-- 1..1 uri|Reference(Any)|Identifier Identity of entity --></what[x]>
  <agent><!-- 0..* Content as for Provenance.agent Entity is attributed to this agent --></agent>
 </entity>
 <signature><!-- 0..* Signature Signature on target --></signature>
</Provenance>
```

JSON Template

```
{
  "resourceType" : "Provenance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "target" : [{ Reference(Any) }], // R!  Target Reference(s) (usually version specific)
  "period" : { Period }, // When the activity occurred
  "recorded" : "<instant>", // R!  When the activity was recorded / updated
  "policy" : ["<uri>"], // Policy or plan the activity was defined by
  "location" : { Reference(Location) }, // Where the activity occurred, if relevant
  "reason" : [{ Coding }], // Reason the activity is occurring
  "activity" : { Coding }, // Activity that occurred
  "agent" : [{ // R!  Actor involved
    "role" : [{ CodeableConcept }], // What the agents role was
    // who[x]: Who participated. One of these 2:
```

```
    "whoUri" : "<uri>",
    "whoReference" : { Reference(Practitioner|RelatedPerson|Patient|Device|
    Organization) },
    // onBehalfOf[x]: Who the agent is representing. One of these 2:
    "onBehalfOfUri" : "<uri>",
    "onBehalfOfReference" : { Reference(Practitioner|RelatedPerson|Patient|Device|
    Organization) },
    "relatedAgentType" : { CodeableConcept } // Type of relationship between agents
  }],
  "entity" : [{ // An entity used in this activity
    "role" : "<code>", // R!  derivation | revision | quotation | source | removal
    // what[x]: Identity of entity. One of these 3:
    "whatUri" : "<uri>",
    "whatReference" : { Reference(Any) },
    "whatIdentifier" : { Identifier },
    "agent" : [{ Content as for Provenance.agent }] // Entity is attributed to this agent
  }],
  "signature" : [{ Signature }] // Signature on target
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Provenance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Provenance.target [ Reference(Any) ], ... ; # 1..* Target Reference(s) (usually version specific)
  fhir:Provenance.period [ Period ]; # 0..1 When the activity occurred
  fhir:Provenance.recorded [ instant ]; # 1..1 When the activity was recorded / updated
  fhir:Provenance.policy [ uri ], ... ; # 0..* Policy or plan the activity was defined by
  fhir:Provenance.location [ Reference(Location) ]; # 0..1 Where the activity occurred, if relevant
  fhir:Provenance.reason [ Coding ], ... ; # 0..* Reason the activity is occurring
  fhir:Provenance.activity [ Coding ]; # 0..1 Activity that occurred
  fhir:Provenance.agent [ # 1..* Actor involved
    fhir:Provenance.agent.role [ CodeableConcept ], ... ; # 0..* What the agents role was
    # Provenance.agent.who[x] : 1..1 Who participated. One of these 2
      fhir:Provenance.agent.whoUri [ uri ]
      fhir:Provenance.agent.whoReference [ Reference(Practitioner|RelatedPerson|Patient|Device|Organization) ]
    # Provenance.agent.onBehalfOf[x] : 0..1 Who the agent is representing. One of these 2
      fhir:Provenance.agent.onBehalfOfUri [ uri ]
      fhir:Provenance.agent.onBehalfOfReference [ Reference(Practitioner|RelatedPerson|Patient|Device|Organization) ]
    fhir:Provenance.agent.relatedAgentType [ CodeableConcept ]; # 0..1 Type of relationship between agents
  ], ...;
  fhir:Provenance.entity [ # 0..* An entity used in this activity
    fhir:Provenance.entity.role [ code ]; # 1..1 derivation | revision | quotation | source | removal
    # Provenance.entity.what[x] : 1..1 Identity of entity. One of these 3
      fhir:Provenance.entity.whatUri [ uri ]
      fhir:Provenance.entity.whatReference [ Reference(Any) ]
      fhir:Provenance.entity.whatIdentifier [ Identifier ]
    fhir:Provenance.entity.agent [ See Provenance.agent ], ... ; # 0..* Entity is attributed to this agent
  ], ...;
  fhir:Provenance.signature [ Signature ], ... ; # 0..* Signature on target
]
```

Changes since DSTU2

Provenance
Provenance.reason | Type changed from CodeableConcept to Coding
Provenance.activity | Type changed from CodeableConcept to Coding Change value set from http://hl7.org/fhir/ValueSet/v3-ProvenanceEventCurrentState to http://hl7.org/fhir/ValueSet/provenance-activity-type
Provenance.agent | Min Cardinality changed from 0 to 1
Provenance.agent.role | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to * Type changed from Coding to CodeableConcept Change value set from http://hl7.org/fhir/ValueSet/provenance-agent-role to http://hl7.org/fhir/ValueSet/security-role-type
Provenance.agent.who[x] | Added Element
Provenance.agent.onBehalfOf[x] | Added Element
Provenance.agent.relatedAgentType | Added Element
Provenance.entity.what[x] | Added Element
Provenance.entity.agent | Max Cardinality changed from 1 to *
Provenance.agent.actor | deleted
Provenance.agent.userId | deleted
Provenance.agent.relatedAgent | deleted
Provenance.entity.type | deleted
Provenance.entity.reference | deleted
Provenance.entity.display | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (1 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

6.3.4.1 Terminology Bindings

Path | Definition | Type | Reference
Provenance.reason | The reason the activity took place. | Extensible | PurposeOfUse
Provenance.activity | The activity that took place. | Extensible | ProvenanceActivityType
Provenance.agent.role | The role that a provenance agent played with respect to the activity. | Extensible | SecurityRoleType
Provenance.agent.relatedAgentType | Type of relationship between two provenance agents. | Example | v3 Code System RoleLinkType
Provenance.entity.role | How an entity was used in an activity. | Required | ProvenanceEntityRole

6.3.4.2 Using the Provenance Resource

The Provenance resource identifies information about another resource (the reference element). The Provenance resource may be used in several different ways:

* As part of a document bundle where it identifies the provenance of part or all of the document

* On a RESTful system where it keeps track of provenance information relating to resources

When used in a document bundle, the references are often not explicitly versioned, but they always implicitly pertain to the version of the resource found in the document. On a RESTful system, the target resource reference should be version specific, but this requires special care: For new resources that need to have a corresponding Provenance resource, the version-specific reference is often not knowable until after the target resource has been updated. This can create an integrity problem for the system - what if the Provenance resource cannot be created after the target resource has been updated? To avoid any such integrity problems, the target resource and the Provenance resources should be submitted as a pair using a transaction.

6.3.4.3 Digital Signatures

The Provenance resource includes a signature element (digital signature) which can be used for standards based integrity verification and non-repudiation purposes. The Signature datatype provides details on use of the signature element. The Signature.type coded value of "Source" should be used when the signature is for simply proving that the resource content is the same as it was when the resource was updated or created.

6.3.4.4 Party References

Because the Provenance resource often refers to parties that are not represented as FHIR resources, agent and entity references can be either references to other resources, or they can refer to other entities that are not FHIR resources.

For Provenance.agent, the actor element is used to reference an existing resource. To reference an entity that is not a FHIR resource, the userId element is used.

A version specific reference to a FHIR resource on the same server:

```
<agent>
  <actor>
    <reference value="Patient/34/_history/3"/>
  </actor>
</agent>
```

A reference to a user (a person) not represented by a FHIR resource:

```
<agent>
  <userId>
    <value value="http://acme.com/users/34"/>
  </userId>
</agent>
```

For Provenance.entity, the code in the .type element is used to differentiate between the two cases: if the code is in the system "http://hl7.org/fhir/resource-types", then the reference is to a resource, and the element reference functions exactly the same as in a resource reference.

A version specific reference to a FHIR resource on the same server:

```
<entity>
  <type>
    <system value="http://hl7.org/fhir/resource-types"/>
    <code value="Patient"/>
  </type>
  <reference value="Patient/34/_history/3"/>
</entity>
```

In effect, this is the same pattern as a standard resource reference, but the type becomes extensible to allow referencing other kinds of resources.

A reference to an entity (a person) not represented by a FHIR resource:

```
<entity>
  <type>
    <system value="http://hl7.org/fhir/provenance-participant-type"/>
    <code value="person"/>
  </type>
  <reference value="http://acme.com/users/34"/>
</entity>
```

One subtle issue with the use of the Provenance resource is to differentiate between whether the reference is to the resource itself, or whether the reference is to the real-world thing that the resource represents, e.g. was it the person involved in the activity, or the record of the person. For agents, it should be understood that the reference is to the real-world thing that the

resource represents.

6.3.4.5 Provenance of Removal

A Provenance record can be recorded to indicate who deleted a Resource. If versioning is supported, the version that was deleted is referenced in Provenance.target; if versioning is not supported then Provenance.target contains the non-version reference. Provenance.entity is not used unless there is a business requirement to do so.

6.3.4.6 Use of Provenance to record Import and Transform

Provenance can be used to record activities of an automaton that transforms input. Such as middleware that extracts information from a HL7 v2 message and creates FHIR resources, or middleware that extracts information from an HL7 CDA document and creates FHIR resources, etc. The Provenance in these cases is recording the activity of the middleware.

The middleware in this case would, in addition to creating the target resources, create a Provenance resource that indicates all the target resources (using Provenance.target). The middleware is identified as one of the Provenance.agent elements, with the Provenance.agent.role of assembler.

The middleware may record the source as another Provenance.agent element.

The original content is optionally saved. This might be as a DocumentReference, or Binary. The Provenance.entity would then point at this original content.

The original source might include some form of 'provenance' to cover the history of the original content prior to the import transformation. This original source 'provenance' should be converted into FHIR Provenance records as appropriate.

6.3.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
agent | reference | Who participated | Provenance.agent.who (Practitioner, Organization, Device, Patient, RelatedPerson)
agent-role | token | What the agents role was | Provenance.agent.role
end | date | End time with inclusive boundary, if not ongoing | Provenance.period.end
entity-id | token | Identity of entity | Provenance.entity.what.as(Identifier)
entity-ref | reference | Identity of entity | Provenance.entity.what.as(Reference) (Any)
location | reference | Where the activity occurred, if relevant | Provenance.location (Location)
patient | reference | Target Reference(s) (usually version specific) | Provenance.target (Patient)
recorded | date | When the activity was recorded / updated | Provenance.recorded
signature-type | token | Indication of the reason the entity signed the object(s) | Provenance.signature.type
start | date | Starting time with inclusive boundary | Provenance.period.start
target | reference | Target Reference(s) (usually version specific) | Provenance.target (Any)

6.3.6 Resource Provenance - Examples

URL: https://hl7.org/fhir/STU3/provenance-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.3.6 Resource Provenance - Examples

Security Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Example List:

General Provenance Example | example | XML | JSON | Turtle
Provenance holding a signature | signature | XML | JSON | Turtle
Provenance with signature for a Consent example | consent-signature | XML | JSON | Turtle
Provenance example for Biocompute objects | example-biocompute-object | XML | JSON | Turtle
Provenance example for cwl workflow project | example-cwl | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

6.3.7 Resource Provenance - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/provenance-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.3.7 Resource Provenance - Detailed Descriptions

Security Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Provenance resource.

Provenance
Definition | Provenance of a resource is a record that describes entities and processes involved in producing and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing authenticity, enabling trust, and allowing reproducibility. Provenance

assertions are a form of contextual metadata and can themselves become important records with their own provenance. Provenance statement indicates clinical significance in terms of confidence in authenticity, reliability, and trustworthiness, integrity, and stage in lifecycle (e.g. Document Completion - has the artifact been legally authenticated), all of which may impact security, privacy, and trust policies.
Control | 1..1
Alternate Names | History; Event; Activity
Comments | Some parties may be duplicated between the target resource and its provenance. For instance, the prescriber is usually (but not always) the author of the prescription resource. This resource is defined with close consideration for W3C Provenance.
Provenance.target
Definition | The Reference(s) that were generated or updated by the activity described in this resource. A provenance can point to more than one target if multiple resources were created/updated by the same activity.
Control | 1..*
Type | Reference(Any)
Summary | true
Comments | Target references are usually version specific, but may not be, if a version has not been assigned or if the provenance information is part of the set of resources being maintained (i.e. a document). When using the RESTful API, the identity of the resource may not be known (especially not the version specific one); the client may either submit the resource first, and then the provenance, or it may submit both using a single transaction. See the notes on transaction for further discussion.
Provenance.period
Definition | The period during which the activity occurred.
Control | 0..1
Type | Period
Comments | The period can be a little arbitrary; where possible, the time should correspond to human assessment of the activity time.
Provenance.recorded
Definition | The instant of time at which the activity was recorded.
Control | 1..1
Type | instant
Summary | true
Comments | This can be a little different from the time stamp on the resource if there is a delay between recording the event and updating the provenance and target resource.
Provenance.policy
Definition | Policy or plan the activity was defined by. Typically, a single activity may have multiple applicable policy documents, such as patient consent, guarantor funding, etc.
Control | 0..*
Type | uri
Comments | For example: Where an OAuth token authorizes, the unique identifier from the OAuth token is placed into the policy element Where a policy engine (e.g. XACML) holds policy logic, the unique policy identifier is placed into the policy element.
Provenance.location
Definition | Where the activity occurred, if relevant.
Control | 0..1
Type | Reference(Location)
Provenance.reason
Definition | The reason that the activity was taking place.
Control | 0..*
Terminology Binding | PurposeOfUse (Extensible)
Type | Coding
Provenance.activity
Definition | An activity is something that occurs over a period of time and acts upon or with entities; it may include consuming, processing, transforming, modifying, relocating, using, or generating entities.
Control | 0..1
Terminology Binding | ProvenanceActivityType (Extensible)
Type | Coding
Provenance.agent
Definition | An actor taking a role in an activity for which it can be assigned some degree of responsibility for the activity taking place.
Control | 1..*
Requirements | An agent can be a person, an organization, software, device, or other entities that may be ascribed responsibility.
Comments | Several agents may be associated (i.e. has some responsibility for an activity) with an activity and vice-versa.
Provenance.agent.role
Definition | The function of the agent with respect to the activity. The security role enabling the agent with respect to the activity.
Control | 0..*
Terminology Binding | SecurityRoleType (Extensible)
Type | CodeableConcept
Summary | true
Comments | For example: author, performer, enterer, attester, doctor, nurse, clerk, etc.
Provenance.agent.who[x]
Definition | The individual, device or organization that participated in the event.
Control | 1..1
Type | uri|Reference(Practitioner | RelatedPerson | Patient | Device | Organization)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Provenance.agent.onBehalfOf[x]
Definition | The individual, device, or organization for whom the change was made.
Control | 0..1
Type | uri|Reference(Practitioner | RelatedPerson | Patient | Device | Organization)

[x] Note | See Choice of Data Types for further information about how to use [x]
Provenance.agent.relatedAgentType
Definition | The type of relationship between agents.
Control | 0..1
Terminology Binding | v3 Code System RoleLinkType (Example)
Type | CodeableConcept
Provenance.entity
Definition | An entity used in this activity.
Control | 0..*
Comments | Multiple userIds may be associated with the same Practitioner or other individual across various appearances, each with distinct privileges.
Provenance.entity.role
Definition | How the entity was used during the activity.
Control | 1..1
Terminology Binding | ProvenanceEntityRole (Required)
Type | code
Summary | true
Provenance.entity.what[x]
Definition | Identity of the Entity used. May be a logical or physical uri and maybe absolute or relative.
Control | 1..1
Type | uri|Reference(Any)|Identifier
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Identity may be a reference to a resource or to something else, depending on the type.
Provenance.entity.agent
Definition | The entity is attributed to an agent to express the agent's responsibility for that entity, possibly along with other agents. This description can be understood as shorthand for saying that the agent was responsible for the activity which generated the entity.
Control | 0..*
Type | See Provenance.agent
Comments | A usecase where one Provenance.entity.agent is used where the Entity that was used in the creation/updating of the Target, is not in the context of the same custodianship as the Target, and thus the meaning of Provenance.entity.agent is to say that the entity referenced is managed elsewhere and that this Agent provided access to it. This would be similar to where the Entity being referenced is managed outside FHIR, such as through HL7 v2, v3, or XDS. This might be where the Entity being referenced is managed in another FHIR resource server. Thus it explains the Provenance of that Entity's use in the context of this Provenance activity.
Provenance.signature
Definition | A digital signature on the target Reference(s). The signer should match a Provenance.agent. The purpose of the signature is indicated.
Control | 0..*
Type | Signature

6.3.8 Resource Provenance - Mappings

URL: https://hl7.org/fhir/STU3/provenance-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.3.8 Resource Provenance - Mappings

Security Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

W3C PROV

FHIR AuditEvent

Mappings for the provenance resource.

6.3.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

Provenance | ControlAct[isNormalAct() and subsumes(CACT, classCode) and moodCode=EVN]
target | ./outboundRelationship[isNormalActRelationship() and typeCode=SUBJ]/target OR ./participation[isNormalParticipation() and typeCode=SBJ]/role OR ./participation[isNormalParticipation() and typeCode=SBJ]/role[isNormalRole()]/player
period | ./effectiveTime[type=IVL_TS]
recorded | unique(./participation[isNormalParticipation() and typeCode=AUT]/time[type=TS])
policy | ./inboundRelationship[isNormalActRelationship() and typeCode="SUBJ"]/source[isNormalAct and subsumes(POLICY, classCode) and moodCode=EVN]/text[typeCode='ED'/tel
location | unique(./participation[isNormalParticipation() and typeCode=LOC]/role[isNormalRole() and subsumes(SDLOC, classCode)]/player[isNormalEntity and classCode="LOC" and determinerCode="INST"]
reason | unique(./reasonCode)
activity | Act.code
agent | ./participation[isNormalParticipation()] OR ./outboundRelationship[isNormalActRelationship() and typeCode='DRIV']
role | .role and/or .typecode
who[x] | .id
onBehalfOf[x] | Person, Practitioner, Organization, Device :* .role [classCode = RoleClassMutualRelationship; role.code and * .scopes[Role](classCode=IDENT) and *.plays [Role.Code]

relatedAgentType | .typeCode
entity | ./subjectOf
role | ./typeCode
what[x] | ./text/reference
agent | ./author/role
signature | ./signatureText

6.3.8.2 Mappings for W3C PROV (http://www.w3.org/ns/prov)

The provenance resource is based on known practices in the HL7 implementation space, particularly those found in the v2 EVN segment, the v3 ControlAct Wrapper, the CDA header, and IHE ATNA. The conceptual model underlying the design is the W3C provenance Specification . Though the content and format of the resource is designed to meet specific requirements for FHIR, all the parts of the resource are formally mapped to the PROV-O specification, and FHIR resources can be transformed to their W3C PROV equivalent.

Provenance | Activity
target | Entity Created/Updated
period | Activity.startTime & Activity.endTime
recorded | Activity.when
policy
location | Activity.location
reason | Activity.Activity
activity | Activity.Activity
agent | Agent
role | Agent.Attribution
who[x]
onBehalfOf[x]
relatedAgentType | type
entity | Entity
role | Entity.role
what[x] | Entity.Identity
agent
signature

6.3.8.3 Mappings for FHIR AuditEvent (http://hl7.org/fhir/auditevent)

Provenance
target | AuditEvent.entity.reference
period
recorded | AuditEvent.recorded
policy | AuditEvent.agent.policy
location | AuditEvent.agent.location
reason | AuditEvent.purposeOfEvent
activity | AuditEvent.purposeOfEvent
agent | AuditEvent.agent
role | AuditEvent.agent.role
who[x]
onBehalfOf[x]
relatedAgentType
entity | AuditEvent.entity
role | AuditEvent.entity.lifecycle
what[x] | AuditEvent.entity.reference
agent
signature

6.3.9 Resource Provenance - Design Notes

URL: https://hl7.org/fhir/STU3/provenance-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.3.9 Resource Provenance - Design Notes

to do

6.3.10 Resource Provenance - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/provenance-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.3.10 Resource Provenance - Extensions & Profiles

Security Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Provenance resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

6.3.10.1 Profiles

ehrsrle-provenance | Defines the elements to be supported within the Provenance resource in order to conform with the Electronic Health Record System Functional Model Record Lifecycle Event standard | for EHRS FM Record Lifecycle Event - Provenance

6.3.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

communicationrequest-relevantHistory | relevantHistory | for CommunicationRequest HL7 Extensions

6.3.10.3 Search Extensions

No Search Extensions defined for this resource

6.3.11 Resource Provenance - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/provenance-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.3.11 Resource Provenance - Detailed Descriptions

Security Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Provenance.

Functional status for this map: 2 tests that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (1 errors). (see documentation)

6.3.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Provenance2to3" = "R2 to R3 Conversions for Provenance"


uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Provenance" alias  ProvenanceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Provenance" alias  Provenance as target

imports "http://hl7.org/fhir/StructureMap/*2to3"


group for type+types  Provenance extends DomainResource
  input src : ProvenanceR2 as source
  input tgt : Provenance as target

  "Provenance-target" : for src.target make tgt.target
  "Provenance-period" : for src.period make tgt.period
  "Provenance-recorded" : for src.recorded make tgt.recorded
  "Provenance-activity" : for src.reason make tgt.reason
  "Provenance-activity" : for src.activity make tgt.activity
  "Provenance-location" : for src.location make tgt.location
  "Provenance-policy" : for src.policy make tgt.policy
  "Provenance-agent" : for src.agent as vs0 make tgt.agent as vt0 then prAgent(vs0, vt0)
  "Provenance-entity" : for src.entity as vs0 make tgt.entity as vt0 then prEntity(vs0, vt0)
  "Provenance-signature" : for src.signature make tgt.signature
endgroup

group prAgent extends BackboneElement
  input src as source
  input tgt as target

  "Provenance.agent-role" : for src.role make tgt.role
  "Provenance.agent-who" : for src.actor make tgt.who
  "Provenance.agent-userId" : for src.userId as su make tgt.who as tu then {
    "Provenance.agent-userId2" : for su make tu.identifier as tu then Identifier(su, tu)
  }
  // related agent ??
endgroup

group prEntity extends BackboneElement
  input src as source
  input tgt as target

  "Provenance.entity-role" : for src.role make tgt.role
  // "Provenance.entity-type" : for src.type make ...?
  // "Provenance.entity-role" : for src.reference make ...?
  // "Provenance.entity-role" : for src.display make ...?
  "Provenance.entity-agent" : for src.agent as vs0 make tgt.agent as vt0 then prAgent(vs0, vt0)
endgroup
```

6.3.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Provenance3to2" = "R3 to R2 Conversion for Provenance"

uses "http://hl7.org/fhir/StructureDefinition/Provenance" alias Provenance as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Provenance" alias ProvenanceR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Provenance extends DomainResource
  input src : Provenance as source
  input tgt : ProvenanceR2 as target

  "Provenance-target" : for src.target make tgt.target
  "Provenance-period" : for src.period make tgt.period
  "Provenance-recorded" : for src.recorded make tgt.recorded
  "Provenance-activity" : for src.reason make tgt.reason
  "Provenance-activity" : for src.activity make tgt.activity
  "Provenance-location" : for src.location make tgt.location
  "Provenance-policy" : for src.policy make tgt.policy
  "Provenance-agent" : for src.agent as vs0 make tgt.agent as vt0 then prAgent(vs0, vt0)
  "Provenance-entity" : for src.entity as vs0 make tgt.entity as vt0 then prEntity(vs0, vt0)
  "Provenance-signature" : for src.signature make tgt.signature
endgroup

group prAgent extends BackboneElement
  input src as source
  input tgt as target

  "Provenance.agent-role" : for src.role make tgt.role
  "Provenance.agent-who" : for src.who where display.exists() or reference.exists() make tgt.actor
  "Provenance.agent-userId" : for src.who as su where identifier.exists() then {
    "Provenance.agent-userId2" : for su.identifier make tgt.userId
  }
  // related agent ??
endgroup

group prEntity extends BackboneElement
  input src as source
  input tgt as target

  "Provenance.entity-role" : for src.role make tgt.role
  // "Provenance.entity-type" : for src.type make ...?
  // "Provenance.entity-role" : for src.reference make ...?
  // "Provenance.entity-role" : for src.display make ...?
  "Provenance.entity-agent" : for src.agent as vs0 make tgt.agent as vt0 then prAgent(vs0, vt0)
endgroup
```

6.3.11.3 R3 Validation Errors

example | ERROR @ Provenance.entity: Profile http://hl7.org/fhir/StructureDefinition/Provenance, Element 'Provenance.entity.what[x]': minimum required = 1, but only found 0 (src = InstanceValidator)

6.4 Resource AuditEvent - Content

URL: https://hl7.org/fhir/STU3/auditevent.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Security Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner

A record of an event made for purposes of maintaining a security log. Typical uses include detection of intrusion attempts and monitoring for inappropriate usage.

6.4.1 Scope and Usage

The audit event is based on the IHE-ATNA Audit record definitions, originally from RFC 3881 , and now managed by DICOM (see DICOM Part 15 Annex A5).

* ASTM E2147 – Setup the concept of security audit logs for healthcare including accounting of disclosures

* IETF RFC 3881 – Defined the Information Model (IETF rule forced this to be informative)

* DICOM Audit Log Message – Made the information model Normative, defined Vocabulary, Transport Binding, and Schema

* IHE ATNA – Defines the grouping with secure transport and access controls; and defined specific audit log records for specific IHE transactions.

* NIST SP800-92 – Shows how to do audit log management and reporting – consistent with our model

* HL7 PASS – Defined an Audit Service with responsibilities and a query interface for reporting use

* ISO 27789 – Defined the subset of audit events that an EHR would need

* ISO/HL7 10781 EHR System Functional Model Release 2

* ISO 21089 Trusted End-to-End Information Flows

This resource is managed collaboratively between HL7, DICOM, and IHE.

The primary purpose of this resource is the maintenance of security audit log information. However, it can also be used for any audit logging needs and simple event-based notification.

6.4.2 Background and Context

All actors - such as applications, processes, and services - involved in an auditable event should record an AuditEvent. This will likely result in multiple AuditEvent entries that show whether privacy and security safeguards, such as access control, are the properly functioning across an enterprise's system-of-systems. Thus, it is typical to get an auditable event recorded by both the application in a workflow process, and the servers that support them. For this reason, duplicate entries are expected, which is helpful because it may aid in the detecting of, for example, fewer than expected actors being recorded in a multi-actor process or attributes related to those records being in conflict, which is an indication of a security problem. There may be non-participating actors that also detect a security relevant event and thus would record an AuditEvent, such as a trusted intermediary.

Security relevant events are not limited to communications or RESTful events. They include: software start-up and shutdown; user login and logout; access control decisions; configuration events; software installation; policy rules changes; and manipulation of data that exposes the data to users. See the Audit Event Sub-Type vocabulary for guidance on some security relevant events.

The content of an AuditEvent is intended for use by Security System Administrators, Security and Privacy Information Managers, and Records Management personnel. This content is not intended to be accessible or used directly by other healthcare users, such as Providers or Patients, although reports generated from the raw data would be useful. An example is a Patient-centric Accounting of Disclosures or an Access Report. Servers that provide support for Audit Event resources would not generally accept update or delete operations on the resources, as this would compromise the integrity of the audit record. Access to the AuditEvent would typically be limited to security, privacy, or other system administration purposes.

Relationship of AuditEvent and Provenance resources are often (though not exclusively) created by the application responding to the create/read/query/update/delete/execute etc. event. A Provenance resource contains overlapping information, but is a record-keeping assertion that gathers information about the context in which the information in a resource "came to be" in its current state, e.g., whether it was created de novo or obtained from another entity in whole, part, or by transformation. Provenance resources are prepared by the application that initiates the create/update of the resource and may be persisted with the AuditEvent target resource.

6.4.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
AuditEvent | | | DomainResource | Event record kept for security purposes Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
type | ? | 1..1 | Coding | Type/identifier of event Audit Event ID (Extensible)
subtype | ? | 0..* | Coding | More specific type/id for the event Audit Event Sub-Type (Extensible)
action | ? | 0..1 | code | Type of action performed during the event AuditEventAction (Required)
recorded | ? | 1..1 | instant | Time when the event occurred on source
outcome | ? | 0..1 | code | Whether the event succeeded or failed AuditEventOutcome (Required)
outcomeDesc | ? | 0..1 | string | Description of the event outcome
purposeOfEvent | ? | 0..* | CodeableConcept | The purposeOfUse of the event PurposeOfUse (Extensible)
agent | | 1..* | BackboneElement | Actor involved in the event
role | | 0..* | CodeableConcept | Agent role in the event SecurityRoleType (Extensible)
reference | ? | 0..1 | Reference(Practitioner | Organization | Device | Patient | RelatedPerson) | Direct reference to resource
userId | ? | 0..1 | Identifier | Unique identifier for the user
altId | | 0..1 | string | Alternative User id e.g. authentication
name | | 0..1 | string | Human-meaningful name for the agent
requestor | | 1..1 | boolean | Whether user is initiator
location | | 0..1 | Reference(Location) | Where
policy | | 0..* | uri | Policy that authorized event
media | | 0..1 | Coding | Type of media Media Type Code (Extensible)
network | | 0..1 | BackboneElement | Logical network location for application activity
address | | 0..1 | string | Identifier for the network access point of the user device
type | | 0..1 | code | The type of network access point AuditEventAgentNetworkType (Required)
purposeOfUse | | 0..* | CodeableConcept | Reason given for this user PurposeOfUse (Extensible)
source | | 1..1 | BackboneElement | Audit Event Reporter
site | | 0..1 | string | Logical source location within the enterprise
identifier | ? | 1..1 | Identifier | The identity of source detecting the event

type | | 0..* | Coding | The type of source where event originated Audit Event Source Type (Extensible)
entity | I | 0..* | BackboneElement | Data or objects used + Either a name or a query (NOT both)
identifier | ? | 0..1 | Identifier | Specific instance of object
reference | ? | 0..1 | Reference(Any) | Specific instance of resource
type | | 0..1 | Coding | Type of entity involved AuditEventEntityType (Extensible)
role | | 0..1 | Coding | What role the entity played AuditEventEntityRole (Extensible)
lifecycle | | 0..1 | Coding | Life-cycle stage for the entity ObjectLifecycleEvents (Extensible)
securityLabel | | 0..* | Coding | Security labels on the entity All Security Labels (Extensible)
name | ?I | 0..1 | string | Descriptor for entity
description | | 0..1 | string | Descriptive text
query | ?I | 0..1 | base64Binary | Query parameters
detail | | 0..* | BackboneElement | Additional Information about the entity
type | | 1..1 | string | Name of the property
value | | 1..1 | base64Binary | Property value
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AuditEvent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <type><!-- 1..1 Coding Type/identifier of event --></type>
 <subtype><!-- 0..* Coding More specific type/id for the event --></subtype>
 <action value="[code]"/><!-- 0..1 Type of action performed during the event -->
 <recorded value="[instant]"/><!-- 1..1 Time when the event occurred on source -->
 <outcome value="[code]"/><!-- 0..1 Whether the event succeeded or failed -->
 <outcomeDesc value="[string]"/><!-- 0..1 Description of the event outcome -->
 <purposeOfEvent><!-- 0..* CodeableConcept The purposeOfUse of the event --></purposeOfEvent>
 <agent>  <!-- 1..* Actor involved in the event -->
  <role><!-- 0..* CodeableConcept Agent role in the event --></role>
  <reference><!-- 0..1 Reference(Practitioner|Organization|Device|Patient|
    RelatedPerson) Direct reference to resource --></reference>
  <userId><!-- 0..1 Identifier Unique identifier for the user --></userId>
  <altId value="[string]"/><!-- 0..1 Alternative User id e.g. authentication -->
  <name value="[string]"/><!-- 0..1 Human-meaningful name for the agent -->
  <requestor value="[boolean]"/><!-- 1..1 Whether user is initiator -->
  <location><!-- 0..1 Reference(Location) Where --></location>
  <policy value="[uri]"/><!-- 0..* Policy that authorized event -->
  <media><!-- 0..1 Coding Type of media --></media>
  <network>  <!-- 0..1 Logical network location for application activity -->
   <address value="[string]"/><!-- 0..1 Identifier for the network access point of the user device -->
   <type value="[code]"/><!-- 0..1 The type of network access point -->
  </network>
  <purposeOfUse><!-- 0..* CodeableConcept Reason given for this user --></purposeOfUse>
 </agent>
 <source>  <!-- 1..1 Audit Event Reporter -->
  <site value="[string]"/><!-- 0..1 Logical source location within the enterprise -->
  <identifier><!-- 1..1 Identifier The identity of source detecting the event --></identifier>
  <type><!-- 0..* Coding The type of source where event originated --></type>
 </source>
 <entity>  <!-- 0..* Data or objects used -->
  <identifier><!-- 0..1 Identifier Specific instance of object --></identifier>
  <reference><!-- 0..1 Reference(Any) Specific instance of resource --></reference>
  <type><!-- 0..1 Coding Type of entity involved --></type>
  <role><!-- 0..1 Coding What role the entity played --></role>
  <lifecycle><!-- 0..1 Coding Life-cycle stage for the entity --></lifecycle>
  <securityLabel><!-- 0..* Coding Security labels on the entity --></securityLabel>
  <name value="[string]"/><!--  0..1 Descriptor for entity -->
  <description value="[string]"/><!-- 0..1 Descriptive text -->
  <query value="[base64Binary]"/><!--  0..1 Query parameters -->
  <detail>  <!-- 0..* Additional Information about the entity -->
   <type value="[string]"/><!-- 1..1 Name of the property -->
   <value value="[base64Binary]"/><!-- 1..1 Property value -->
  </detail>
 </entity>
</AuditEvent>
```

JSON Template

```
{
  "resourceType" : "AuditEvent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "type" : { Coding }, // R!  Type/identifier of event
  "subtype" : [{ Coding }], // More specific type/id for the event
  "action" : "<code>", // Type of action performed during the event
  "recorded" : "<instant>", // R!  Time when the event occurred on source
  "outcome" : "<code>", // Whether the event succeeded or failed
  "outcomeDesc" : "<string>", // Description of the event outcome
  "purposeOfEvent" : [{ CodeableConcept }], // The purposeOfUse of the event
  "agent" : [{ // R!  Actor involved in the event
    "role" : [{ CodeableConcept }], // Agent role in the event
    "reference" : { Reference(Practitioner|Organization|Device|Patient|
    RelatedPerson) }, // Direct reference to resource
    "userId" : { Identifier }, // Unique identifier for the user
    "altId" : "<string>", // Alternative User id e.g. authentication
```

```
    "name" : "<string>", // Human-meaningful name for the agent
    "requestor" : <boolean>, // R!  Whether user is initiator
    "location" : { Reference(Location) }, // Where
    "policy" : ["<uri>"], // Policy that authorized event
    "media" : { Coding }, // Type of media
    "network" : { // Logical network location for application activity
      "address" : "<string>", // Identifier for the network access point of the user device
      "type" : "<code>" // The type of network access point
    },
    "purposeOfUse" : [{ CodeableConcept }] // Reason given for this user
  }],
  "source" : { // R!  Audit Event Reporter
    "site" : "<string>", // Logical source location within the enterprise
    "identifier" : { Identifier }, // R!  The identity of source detecting the event
    "type" : [{ Coding }] // The type of source where event originated
  },
  "entity" : [{ // Data or objects used
    "identifier" : { Identifier }, // Specific instance of object
    "reference" : { Reference(Any) }, // Specific instance of resource
    "type" : { Coding }, // Type of entity involved
    "role" : { Coding }, // What role the entity played
    "lifecycle" : { Coding }, // Life-cycle stage for the entity
    "securityLabel" : [{ Coding }], // Security labels on the entity
    "name" : "<string>", // C? Descriptor for entity
    "description" : "<string>", // Descriptive text
    "query" : "<base64Binary>", // C? Query parameters
    "detail" : [{ // Additional Information about the entity
      "type" : "<string>", // R!  Name of the property
      "value" : "<base64Binary>" // R!  Property value
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AuditEvent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AuditEvent.type [ Coding ]; # 1..1 Type/identifier of event
  fhir:AuditEvent.subtype [ Coding ], ... ; # 0..* More specific type/id for the event
  fhir:AuditEvent.action [ code ]; # 0..1 Type of action performed during the event
  fhir:AuditEvent.recorded [ instant ]; # 1..1 Time when the event occurred on source
  fhir:AuditEvent.outcome [ code ]; # 0..1 Whether the event succeeded or failed
  fhir:AuditEvent.outcomeDesc [ string ]; # 0..1 Description of the event outcome
  fhir:AuditEvent.purposeOfEvent [ CodeableConcept ], ... ; # 0..* The purposeOfUse of the event
  fhir:AuditEvent.agent [ # 1..* Actor involved in the event
    fhir:AuditEvent.agent.role [ CodeableConcept ], ... ; # 0..* Agent role in the event
    fhir:AuditEvent.agent.reference [ Reference(Practitioner|Organization|Device|Patient|RelatedPerson) ]; # 0..1 Direct
     reference to resource
    fhir:AuditEvent.agent.userId [ Identifier ]; # 0..1 Unique identifier for the user
    fhir:AuditEvent.agent.altId [ string ]; # 0..1 Alternative User id e.g. authentication
    fhir:AuditEvent.agent.name [ string ]; # 0..1 Human-meaningful name for the agent
    fhir:AuditEvent.agent.requestor [ boolean ]; # 1..1 Whether user is initiator
    fhir:AuditEvent.agent.location [ Reference(Location) ]; # 0..1 Where
    fhir:AuditEvent.agent.policy [ uri ], ... ; # 0..* Policy that authorized event
    fhir:AuditEvent.agent.media [ Coding ]; # 0..1 Type of media
    fhir:AuditEvent.agent.network [ # 0..1 Logical network location for application activity
      fhir:AuditEvent.agent.network.address [ string ]; # 0..1 Identifier for the network access point of the user
    device
      fhir:AuditEvent.agent.network.type [ code ]; # 0..1 The type of network access point
    ];
    fhir:AuditEvent.agent.purposeOfUse [ CodeableConcept ], ... ; # 0..* Reason given for this user
  ], ...;
  fhir:AuditEvent.source [ # 1..1 Audit Event Reporter
    fhir:AuditEvent.source.site [ string ]; # 0..1 Logical source location within the enterprise
    fhir:AuditEvent.source.identifier [ Identifier ]; # 1..1 The identity of source detecting the event
    fhir:AuditEvent.source.type [ Coding ], ... ; # 0..* The type of source where event originated
  ];
  fhir:AuditEvent.entity [ # 0..* Data or objects used
    fhir:AuditEvent.entity.identifier [ Identifier ]; # 0..1 Specific instance of object
    fhir:AuditEvent.entity.reference [ Reference(Any) ]; # 0..1 Specific instance of resource
    fhir:AuditEvent.entity.type [ Coding ]; # 0..1 Type of entity involved
    fhir:AuditEvent.entity.role [ Coding ]; # 0..1 What role the entity played
    fhir:AuditEvent.entity.lifecycle [ Coding ]; # 0..1 Life-cycle stage for the entity
    fhir:AuditEvent.entity.securityLabel [ Coding ], ... ; # 0..* Security labels on the entity
    fhir:AuditEvent.entity.name [ string ]; # 0..1 Descriptor for entity
    fhir:AuditEvent.entity.description [ string ]; # 0..1 Descriptive text
    fhir:AuditEvent.entity.query [ base64Binary ]; # 0..1 Query parameters
    fhir:AuditEvent.entity.detail [ # 0..* Additional Information about the entity
      fhir:AuditEvent.entity.detail.type [ string ]; # 1..1 Name of the property
      fhir:AuditEvent.entity.detail.value [ base64Binary ]; # 1..1 Property value
    ], ...;
  ], ...;
]
```

Changes since DSTU2

AuditEvent
AuditEvent.type | Added Element
AuditEvent.subtype | Added Element
AuditEvent.action | Added Element
AuditEvent.recorded | Added Element
AuditEvent.outcome | Added Element
AuditEvent.outcomeDesc | Added Element
AuditEvent.purposeOfEvent | Added Element
AuditEvent.agent | Renamed from participant to agent
AuditEvent.agent.role | Change value set from http://hl7.org/fhir/ValueSet/dicm-402-roleid to http://hl7.org/fhir/ValueSet/security-role-type
AuditEvent.agent.purposeOfUse | Type changed from Coding to CodeableConcept
AuditEvent.entity | Renamed from object to entity
AuditEvent.entity.type | Change value set from http://hl7.org/fhir/ValueSet/object-type to http://hl7.org/fhir/ValueSet/audit-entity-type
AuditEvent.entity.lifecycle | Change value set from http://hl7.org/fhir/ValueSet/object-lifecycle to http://hl7.org/fhir/ValueSet/object-lifecycle-events
AuditEvent.event | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 8 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
AuditEvent | | | DomainResource | Event record kept for security purposes Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
type | ? | 1..1 | Coding | Type/identifier of event Audit Event ID (Extensible)
subtype | ? | 0..* | Coding | More specific type/id for the event Audit Event Sub-Type (Extensible)
action | ? | 0..1 | code | Type of action performed during the event AuditEventAction (Required)
recorded | ? | 1..1 | instant | Time when the event occurred on source
outcome | ? | 0..1 | code | Whether the event succeeded or failed AuditEventOutcome (Required)
outcomeDesc | ? | 0..1 | string | Description of the event outcome
purposeOfEvent | ? | 0..* | CodeableConcept | The purposeOfUse of the event PurposeOfUse (Extensible)
agent | | 1..* | BackboneElement | Actor involved in the event
role | | 0..* | CodeableConcept | Agent role in the event SecurityRoleType (Extensible)
reference | ? | 0..1 | Reference(Practitioner | Organization | Device | Patient | RelatedPerson) | Direct reference to resource
userId | ? | 0..1 | Identifier | Unique identifier for the user
altId | | 0..1 | string | Alternative User id e.g. authentication
name | | 0..1 | string | Human-meaningful name for the agent
requestor | | 1..1 | boolean | Whether user is initiator
location | | 0..1 | Reference(Location) | Where
policy | | 0..* | uri | Policy that authorized event
media | | 0..1 | Coding | Type of media Media Type Code (Extensible)
network | | 0..1 | BackboneElement | Logical network location for application activity
address | | 0..1 | string | Identifier for the network access point of the user device
type | | 0..1 | code | The type of network access point AuditEventAgentNetworkType (Required)
purposeOfUse | | 0..* | CodeableConcept | Reason given for this user PurposeOfUse (Extensible)
source | | 1..1 | BackboneElement | Audit Event Reporter
site | | 0..1 | string | Logical source location within the enterprise
identifier | ? | 1..1 | Identifier | The identity of source detecting the event
type | | 0..* | Coding | The type of source where event originated Audit Event Source Type (Extensible)
entity | I | 0..* | BackboneElement | Data or objects used + Either a name or a query (NOT both)
identifier | ? | 0..1 | Identifier | Specific instance of object
reference | ? | 0..1 | Reference(Any) | Specific instance of resource
type | | 0..1 | Coding | Type of entity involved AuditEventEntityType (Extensible)
role | | 0..1 | Coding | What role the entity played AuditEventEntityRole (Extensible)
lifecycle | | 0..1 | Coding | Life-cycle stage for the entity ObjectLifecycleEvents (Extensible)
securityLabel | | 0..* | Coding | Security labels on the entity All Security Labels (Extensible)
name | ?I | 0..1 | string | Descriptor for entity
description | | 0..1 | string | Descriptive text
query | ?I | 0..1 | base64Binary | Query parameters
detail | | 0..* | BackboneElement | Additional Information about the entity
type | | 1..1 | string | Name of the property
value | | 1..1 | base64Binary | Property value
Documentation for this format

UML Diagram (Legend)

XML Template

```
<AuditEvent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <type><!-- 1..1 Coding Type/identifier of event --></type>
 <subtype><!-- 0..* Coding More specific type/id for the event --></subtype>
```

```
 <action value="[code]"/><!-- 0..1 Type of action performed during the event -->
 <recorded value="[instant]"/><!-- 1..1 Time when the event occurred on source -->
 <outcome value="[code]"/><!-- 0..1 Whether the event succeeded or failed -->
 <outcomeDesc value="[string]"/><!-- 0..1 Description of the event outcome -->
 <purposeOfEvent><!-- 0..* CodeableConcept The purposeOfUse of the event --></purposeOfEvent>
 <agent>  <!-- 1..* Actor involved in the event -->
  <role><!-- 0..* CodeableConcept Agent role in the event --></role>
  <reference><!-- 0..1 Reference(Practitioner|Organization|Device|Patient|
    RelatedPerson) Direct reference to resource --></reference>
  <userId><!-- 0..1 Identifier Unique identifier for the user --></userId>
  <altId value="[string]"/><!-- 0..1 Alternative User id e.g. authentication -->
  <name value="[string]"/><!-- 0..1 Human-meaningful name for the agent -->
  <requestor value="[boolean]"/><!-- 1..1 Whether user is initiator -->
  <location><!-- 0..1 Reference(Location) Where --></location>
  <policy value="[uri]"/><!-- 0..* Policy that authorized event -->
  <media><!-- 0..1 Coding Type of media --></media>
  <network>  <!-- 0..1 Logical network location for application activity -->
   <address value="[string]"/><!-- 0..1 Identifier for the network access point of the user device -->
   <type value="[code]"/><!-- 0..1 The type of network access point -->
  </network>
  <purposeOfUse><!-- 0..* CodeableConcept Reason given for this user --></purposeOfUse>
 </agent>
 <source>  <!-- 1..1 Audit Event Reporter -->
  <site value="[string]"/><!-- 0..1 Logical source location within the enterprise -->
  <identifier><!-- 1..1 Identifier The identity of source detecting the event --></identifier>
  <type><!-- 0..* Coding The type of source where event originated --></type>
 </source>
 <entity>  <!-- 0..* Data or objects used -->
  <identifier><!-- 0..1 Identifier Specific instance of object --></identifier>
  <reference><!-- 0..1 Reference(Any) Specific instance of resource --></reference>
  <type><!-- 0..1 Coding Type of entity involved --></type>
  <role><!-- 0..1 Coding What role the entity played --></role>
  <lifecycle><!-- 0..1 Coding Life-cycle stage for the entity --></lifecycle>
  <securityLabel><!-- 0..* Coding Security labels on the entity --></securityLabel>
  <name value="[string]"/><!--  0..1 Descriptor for entity -->
  <description value="[string]"/><!-- 0..1 Descriptive text -->
  <query value="[base64Binary]"/><!--  0..1 Query parameters -->
  <detail>  <!-- 0..* Additional Information about the entity -->
   <type value="[string]"/><!-- 1..1 Name of the property -->
   <value value="[base64Binary]"/><!-- 1..1 Property value -->
  </detail>
 </entity>
</AuditEvent>
```

JSON Template

```
{
  "resourceType" : "AuditEvent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "type" : { Coding }, // R!  Type/identifier of event
  "subtype" : [{ Coding }], // More specific type/id for the event
  "action" : "<code>", // Type of action performed during the event
  "recorded" : "<instant>", // R!  Time when the event occurred on source
  "outcome" : "<code>", // Whether the event succeeded or failed
  "outcomeDesc" : "<string>", // Description of the event outcome
  "purposeOfEvent" : [{ CodeableConcept }], // The purposeOfUse of the event
  "agent" : [{ // R!  Actor involved in the event
    "role" : [{ CodeableConcept }], // Agent role in the event
    "reference" : { Reference(Practitioner|Organization|Device|Patient|
    RelatedPerson) }, // Direct reference to resource
    "userId" : { Identifier }, // Unique identifier for the user
    "altId" : "<string>", // Alternative User id e.g. authentication
    "name" : "<string>", // Human-meaningful name for the agent
    "requestor" : <boolean>, // R!  Whether user is initiator
    "location" : { Reference(Location) }, // Where
    "policy" : ["<uri>"], // Policy that authorized event
    "media" : { Coding }, // Type of media
    "network" : { // Logical network location for application activity
      "address" : "<string>", // Identifier for the network access point of the user device
      "type" : "<code>" // The type of network access point
    },
    "purposeOfUse" : [{ CodeableConcept }] // Reason given for this user
  }],
  "source" : { // R!  Audit Event Reporter
    "site" : "<string>", // Logical source location within the enterprise
    "identifier" : { Identifier }, // R!  The identity of source detecting the event
    "type" : [{ Coding }] // The type of source where event originated
  },
  "entity" : [{ // Data or objects used
    "identifier" : { Identifier }, // Specific instance of object
    "reference" : { Reference(Any) }, // Specific instance of resource
    "type" : { Coding }, // Type of entity involved
    "role" : { Coding }, // What role the entity played
    "lifecycle" : { Coding }, // Life-cycle stage for the entity
    "securityLabel" : [{ Coding }], // Security labels on the entity
    "name" : "<string>", // C? Descriptor for entity
    "description" : "<string>", // Descriptive text
    "query" : "<base64Binary>", // C? Query parameters
    "detail" : [{ // Additional Information about the entity
```

```
      "type" : "<string>", // R!  Name of the property
      "value" : "<base64Binary>" // R!  Property value
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:AuditEvent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:AuditEvent.type [ Coding ]; # 1..1 Type/identifier of event
  fhir:AuditEvent.subtype [ Coding ], ... ; # 0..* More specific type/id for the event
  fhir:AuditEvent.action [ code ]; # 0..1 Type of action performed during the event
  fhir:AuditEvent.recorded [ instant ]; # 1..1 Time when the event occurred on source
  fhir:AuditEvent.outcome [ code ]; # 0..1 Whether the event succeeded or failed
  fhir:AuditEvent.outcomeDesc [ string ]; # 0..1 Description of the event outcome
  fhir:AuditEvent.purposeOfEvent [ CodeableConcept ], ... ; # 0..* The purposeOfUse of the event
  fhir:AuditEvent.agent [ # 1..* Actor involved in the event
    fhir:AuditEvent.agent.role [ CodeableConcept ], ... ; # 0..* Agent role in the event
    fhir:AuditEvent.agent.reference [ Reference(Practitioner|Organization|Device|Patient|RelatedPerson) ]; # 0..1 Direct
     reference to resource
    fhir:AuditEvent.agent.userId [ Identifier ]; # 0..1 Unique identifier for the user
    fhir:AuditEvent.agent.altId [ string ]; # 0..1 Alternative User id e.g. authentication
    fhir:AuditEvent.agent.name [ string ]; # 0..1 Human-meaningful name for the agent
    fhir:AuditEvent.agent.requestor [ boolean ]; # 1..1 Whether user is initiator
    fhir:AuditEvent.agent.location [ Reference(Location) ]; # 0..1 Where
    fhir:AuditEvent.agent.policy [ uri ], ... ; # 0..* Policy that authorized event
    fhir:AuditEvent.agent.media [ Coding ]; # 0..1 Type of media
    fhir:AuditEvent.agent.network [ # 0..1 Logical network location for application activity
      fhir:AuditEvent.agent.network.address [ string ]; # 0..1 Identifier for the network access point of the user
    device
      fhir:AuditEvent.agent.network.type [ code ]; # 0..1 The type of network access point
    ];
    fhir:AuditEvent.agent.purposeOfUse [ CodeableConcept ], ... ; # 0..* Reason given for this user
  ], ...;
  fhir:AuditEvent.source [ # 1..1 Audit Event Reporter
    fhir:AuditEvent.source.site [ string ]; # 0..1 Logical source location within the enterprise
    fhir:AuditEvent.source.identifier [ Identifier ]; # 1..1 The identity of source detecting the event
    fhir:AuditEvent.source.type [ Coding ], ... ; # 0..* The type of source where event originated
  ];
  fhir:AuditEvent.entity [ # 0..* Data or objects used
    fhir:AuditEvent.entity.identifier [ Identifier ]; # 0..1 Specific instance of object
    fhir:AuditEvent.entity.reference [ Reference(Any) ]; # 0..1 Specific instance of resource
    fhir:AuditEvent.entity.type [ Coding ]; # 0..1 Type of entity involved
    fhir:AuditEvent.entity.role [ Coding ]; # 0..1 What role the entity played
    fhir:AuditEvent.entity.lifecycle [ Coding ]; # 0..1 Life-cycle stage for the entity
    fhir:AuditEvent.entity.securityLabel [ Coding ], ... ; # 0..* Security labels on the entity
    fhir:AuditEvent.entity.name [ string ]; # 0..1 Descriptor for entity
    fhir:AuditEvent.entity.description [ string ]; # 0..1 Descriptive text
    fhir:AuditEvent.entity.query [ base64Binary ]; # 0..1 Query parameters
    fhir:AuditEvent.entity.detail [ # 0..* Additional Information about the entity
      fhir:AuditEvent.entity.detail.type [ string ]; # 1..1 Name of the property
      fhir:AuditEvent.entity.detail.value [ base64Binary ]; # 1..1 Property value
    ], ...;
  ], ...;
]
```

Changes since DSTU2

AuditEvent
AuditEvent.type | Added Element
AuditEvent.subtype | Added Element
AuditEvent.action | Added Element
AuditEvent.recorded | Added Element
AuditEvent.outcome | Added Element
AuditEvent.outcomeDesc | Added Element
AuditEvent.purposeOfEvent | Added Element
AuditEvent.agent | Renamed from participant to agent
AuditEvent.agent.role | Change value set from http://hl7.org/fhir/ValueSet/dicm-402-roleid to http://hl7.org/fhir/ValueSet/security-role-type
AuditEvent.agent.purposeOfUse | Type changed from Coding to CodeableConcept
AuditEvent.entity | Renamed from object to entity
AuditEvent.entity.type | Change value set from http://hl7.org/fhir/ValueSet/object-type to http://hl7.org/fhir/ValueSet/audit-entity-type
AuditEvent.entity.lifecycle | Change value set from http://hl7.org/fhir/ValueSet/object-lifecycle to http://hl7.org/fhir/ValueSet/object-lifecycle-events
AuditEvent.event | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 8 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

6.4.3.1 Terminology Bindings

Path | Definition | Type | Reference
AuditEvent.type | Type of event. | Extensible | Audit Event ID
AuditEvent.subtype | Sub-type of event. | Extensible | Audit Event Sub-Type
AuditEvent.action | Indicator for type of action performed during the event that generated the event | Required | AuditEventAction
AuditEvent.outcome | Indicates whether the event succeeded or failed | Required | AuditEventOutcome
AuditEvent.purposeOfEvent AuditEvent.agent.purposeOfUse | The reason the activity took place. | Extensible | PurposeOfUse
AuditEvent.agent.role | What security role enabled the agent to participate in the event | Extensible | SecurityRoleType
AuditEvent.agent.media | Used when the event is about exporting/importing onto media. | Extensible | Media Type Code
AuditEvent.agent.network.type | The type of network access point of this agent in the audit event | Required | AuditEventAgentNetworkType
AuditEvent.source.type | Code specifying the type of system that detected and recorded the event. | Extensible | Audit Event Source Type
AuditEvent.entity.type | Code for the entity type involved in the audit event | Extensible | AuditEventEntityType
AuditEvent.entity.role | Code representing the role the entity played in the audit event | Extensible | AuditEventEntityRole
AuditEvent.entity.lifecycle | Identifier for the data life-cycle stage for the entity | Extensible | ObjectLifecycleEvents
AuditEvent.entity.securityLabel | Security Labels from the Healthcare Privacy and Security Classification System. | Extensible | All Security Labels

6.4.3.2 Constraints

* sev-1: On AuditEvent.entity: Either a name or a query (NOT both) (expression on AuditEvent.entity: name.empty() or query.empty())

6.4.3.3 Using Coded Values

The AuditEvent resource and the ATNA Audit record are used in many contexts throughout healthcare. The coded values defined in the "extensible" bindings above are those widely used and/or defined by DICOM, IHE or ISO, who defined these codes to meet very specific use cases. These codes should be used when they are suitable. When needed, other codes can be defined.

Note: When using codes from a vocabulary, the displayName can be left off to keep the AuditEvent size small and minimize impact of a large audit log of similar entries.

The set of codes defined for this resource is expected to grow over time, and additional codes may be proposed / requested using the "Propose a change" link above below.

6.4.3.4 Event codes for Common Scenarios

This table summarizes common event scenarios, and the codes that should be used for each case.

Scenario | type | subtype | action | Other
User Login (example) | 110114 User Authentication | 110122 User Authentication | E Execute | One participant which contains the details of the logged-in user.
User Logout (example) | 110114 User Authentication | 110123 User Logout | E Execute | One participant which contains the details of the logged-out user.
REST operation logged on server (example) | rest RESTful Operation | [code] defined for operation | * (see below) | Participant for logged in user, if available, and one object with a reference, if at least the type is known as part of the operation. Reference.url should be provided to the granularity known.
Search operation logged on server (example) | rest RESTful Operation | [code] defined for operation | E Execute | Participant for logged in user, if available, and one object with a query element.

Audit Event Actions for RESTful operations:

Operation | Action
create | C
read, vread, history-instance, history-type, history-system | R
update | U
delete | D
transaction, operation, conformance, validate, search, search-type, search-system | E

6.4.3.5 PurposeOfEvent and PurposeOfUse

The audit event provides the element purposeOfEvent to convey the purpose of the event and purposeOfUse to convey the reason that a particular actor (machine, person, software) was involved in the event.

purposeOfEvent is an element at the level of AuditEvent and can convey the purpose of the activity that resulted in the event. This will occur when the system that is reporting the event is be aware of the purpose of the event. A specific example would be a radiology reporting system where a radiologist has created and is sending a finished report. This system likely knows the purpose, e.g., "treatment". It is multi-valued because the one event may be related to multiple purposes.

It is also commonplace that the reporting system does not have information about the purpose of the event. In these cases, the event report would not have a purposeOfEvent.

It is also likely that the same event will be reported from different perspectives, e.g., by both the sender and recipient of a communication. These two different perspectives can have different knowledge regarding the purposeOfEvent.

purposeOfUse is an element at the level of agent within AuditEvent. This describes the reason that this person, machine, or software is participating in the activity that resulted in the event. For example, an individual person participating in the event may assert a purpose of use from their perspective. It is also possible that they are participating for multiple reasons, and report multiple purposeOfUse.

The reporting system might not have knowledge regarding why a particular machine or person was involved, and would omit this element in those cases.

When the same event is reported from multiple perspectives, the reports can have different knowledge regarding the purpose.

6.4.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
action | token | Type of action performed during the event | AuditEvent.action
address | string | Identifier for the network access point of the user device | AuditEvent.agent.network.address
agent | reference | Direct reference to resource | AuditEvent.agent.reference (Practitioner, Organization, Device, Patient, RelatedPerson)
agent-name | string | Human-meaningful name for the agent | AuditEvent.agent.name
agent-role | token | Agent role in the event | AuditEvent.agent.role
altid | token | Alternative User id e.g. authentication | AuditEvent.agent.altId
date | date | Time when the event occurred on source | AuditEvent.recorded
entity | reference | Specific instance of resource | AuditEvent.entity.reference (Any)
entity-id | token | Specific instance of object | AuditEvent.entity.identifier
entity-name | string | Descriptor for entity | AuditEvent.entity.name
entity-role | token | What role the entity played | AuditEvent.entity.role
entity-type | token | Type of entity involved | AuditEvent.entity.type
outcome | token | Whether the event succeeded or failed | AuditEvent.outcome
patient | reference | Direct reference to resource | AuditEvent.agent.reference | AuditEvent.entity.reference (Patient)
policy | uri | Policy that authorized event | AuditEvent.agent.policy
site | token | Logical source location within the enterprise | AuditEvent.source.site
source | token | The identity of source detecting the event | AuditEvent.source.identifier
subtype | token | More specific type/id for the event | AuditEvent.subtype
type | token | Type/identifier of event | AuditEvent.type
user | token | Unique identifier for the user | AuditEvent.agent.userId

6.4.5 Resource AuditEvent - Examples

URL: https://hl7.org/fhir/STU3/auditevent-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.4.5 Resource AuditEvent - Examples

Security Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Example List:

General AuditEvent Example | example | XML | JSON | Turtle
Login example | example-login | XML | JSON | Turtle
Logout example | example-logout | XML | JSON | Turtle
RESTful vread Operation | example-rest | XML | JSON | Turtle
Accounting of a Disclosure | example-disclosure | XML | JSON | Turtle
RESTful search operation | example-search | XML | JSON | Turtle
Converted ATNA message from a PIX query | example-pixQuery | XML | JSON | Turtle
Export of DocumentManifest to Media | example-media | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

6.4.6 Resource AuditEvent - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/auditevent-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.4.6 Resource AuditEvent - Detailed Descriptions

Security Work Group | Maturity Level: 3 | Trial Use | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the AuditEvent resource.

AuditEvent
Definition | A record of an event made for purposes of maintaining a security log. Typical uses include detection of intrusion attempts and monitoring for inappropriate usage.
Control | 1..1
Comments | Based on ATNA (RFC 3881).
AuditEvent.type
Definition | Identifier for a family of the event. For example, a menu item, program, rule, policy, function code, application name or URL. It identifies the performed function.
Control | 1..1

Terminology Binding | Audit Event ID (Extensible)
Type | Coding
Requirements | This identifies the performed function. For "Execute" Event Action Code audit records, this identifies the application function performed.
Summary | true
AuditEvent.subtype
Definition | Identifier for the category of event.
Control | 0..*
Terminology Binding | Audit Event Sub-Type (Extensible)
Type | Coding
Requirements | This field enables queries of messages by implementation-defined event categories.
Summary | true
AuditEvent.action
Definition | Indicator for type of action performed during the event that generated the audit.
Control | 0..1
Terminology Binding | AuditEventAction (Required)
Type | code
Requirements | This broadly indicates what kind of action was done on the AuditEvent.entity by the AuditEvent.agent.
Summary | true
AuditEvent.recorded
Definition | The time when the event occurred on the source.
Control | 1..1
Type | instant
Requirements | This ties an event to a specific date and time. Security audits typically require a consistent time base (e.g. UTC), to eliminate time-zone issues arising from geographical distribution.
Summary | true
Comments | In a distributed system, some sort of common time base (e.g. an NTP [RFC1305] server) is a good implementation tactic.
AuditEvent.outcome
Definition | Indicates whether the event succeeded or failed.
Control | 0..1
Terminology Binding | AuditEventOutcome (Required)
Type | code
Summary | true
Comments | In some cases a "success" may be partial, for example, an incomplete or interrupted transfer of a radiological study. For the purpose of establishing accountability, these distinctions are not relevant.
AuditEvent.outcomeDesc
Definition | A free text description of the outcome of the event.
Control | 0..1
Type | string
Summary | true
AuditEvent.purposeOfEvent
Definition | The purposeOfUse (reason) that was used during the event being recorded.
Control | 0..*
Terminology Binding | PurposeOfUse (Extensible)
Type | CodeableConcept
Summary | true
Comments | Use AuditEvent.agent.purposeOfUse when you know that it is specific to the agent, otherwise use AuditEvent.purposeOfEvent. For example, during a machine-to-machine transfer it might not be obvious to the audit system who caused the event, but it does know why.
AuditEvent.agent
Definition | An actor taking an active role in the event or activity that is logged.
Control | 1..*
Requirements | An agent can be a person, an organization, software, device, or other actors that may be ascribed responsibility.
Alternate Names | ActiveParticipant
Comments | Several agents may be associated (i.e. have some responsibility for an activity) with an event or activity. For example, an activity may be initiated by one user for other users or involve more than one user. However, only one user may be the initiator/requestor for the activity.
AuditEvent.agent.role
Definition | The security role that the user was acting under, that come from local codes defined by the access control security system (e.g. RBAC, ABAC) used in the local context.
Control | 0..*
Terminology Binding | SecurityRoleType (Extensible)
Type | CodeableConcept
Requirements | This value ties an audited event to a user's role(s). It is an optional value that might be used to group events for analysis by user functional role categories.
Comments | Should be roles relevant to the event. Should not be an exhaustive list of roles.
AuditEvent.agent.reference
Definition | Direct reference to a resource that identifies the agent.
Control | 0..1
Type | Reference(Practitioner | Organization | Device | Patient | RelatedPerson)
Requirements | This field ties an audit event to a specific resource.
Summary | true
AuditEvent.agent.userId
Definition | Unique identifier for the user actively participating in the event.
Control | 0..1

Type | Identifier
Requirements | This field ties an audit event to a specific user.
Summary | true
Comments | A unique value within the Audit Source ID. For node-based authentication -- where only the system hardware or process, but not a human user, is identified -- User ID would be the node name.
AuditEvent.agent.altId
Definition | Alternative agent Identifier. For a human, this should be a user identifier text string from authentication system. This identifier would be one known to a common authentication system (e.g. single sign-on), if available.
Control | 0..1
Type | string
Requirements | In some situations, a human user may authenticate with one identity but, to access a specific application system, may use a synonymous identify. For example, some "single sign on" implementations will do this. The alternative identifier would then be the original identify used for authentication, and the User ID is the one known to and used by the application.
AuditEvent.agent.name
Definition | Human-meaningful name for the agent.
Control | 0..1
Type | string
Requirements | The User ID and Authorization User ID may be internal or otherwise obscure values. This field assists the auditor in identifying the actual user.
AuditEvent.agent.requestor
Definition | Indicator that the user is or is not the requestor, or initiator, for the event being audited.
Control | 1..1
Type | boolean
Requirements | This value is used to distinguish between requestor-users and recipient-users. For example, one person may initiate a report-output to be sent to another user.
Comments | There can only be one initiator. If the initiator is not clear, then do not choose any one agent as the initiator.
AuditEvent.agent.location
Definition | Where the event occurred.
Control | 0..1
Type | Reference(Location)
AuditEvent.agent.policy
Definition | The policy or plan that authorized the activity being recorded. Typically, a single activity may have multiple applicable policies, such as patient consent, guarantor funding, etc. The policy would also indicate the security token used.
Control | 0..*
Type | uri
Requirements | This value is used retrospectively to determine the authorization policies.
Comments | For example: Where an OAuth token authorizes, the unique identifier from the OAuth token is placed into the policy element Where a policy engine (e.g. XACML) holds policy logic, the unique policy identifier is placed into the policy element.
AuditEvent.agent.media
Definition | Type of media involved. Used when the event is about exporting/importing onto media.
Control | 0..1
Terminology Binding | Media Type Code (Extensible)
Type | Coding
Requirements | Usually, this is used instead of specifying a network address. This field is not used for Media Id (i.e. the serial number of a CD).
AuditEvent.agent.network
Definition | Logical network location for application activity, if the activity has a network location.
Control | 0..1
AuditEvent.agent.network.address
Definition | An identifier for the network access point of the user device for the audit event.
Control | 0..1
Type | string
Requirements | This datum identifies the user's network access point, which may be distinct from the server that performed the action. It is an optional value that may be used to group events recorded on separate servers for analysis of a specific network access point's data access across all servers.
Comments | This could be a device id, IP address or some other identifier associated with a device.
AuditEvent.agent.network.type
Definition | An identifier for the type of network access point that originated the audit event.
Control | 0..1
Terminology Binding | AuditEventAgentNetworkType (Required)
Type | code
Requirements | This datum identifies the type of network access point identifier of the user device for the audit event. It is an optional value that may be used to group events recorded on separate servers for analysis of access according to a network access point's type.
AuditEvent.agent.purposeOfUse
Definition | The reason (purpose of use), specific to this agent, that was used during the event being recorded.
Control | 0..*
Terminology Binding | PurposeOfUse (Extensible)
Type | CodeableConcept
Comments | Use AuditEvent.agent.purposeOfUse when you know that is specific to the agent, otherwise use AuditEvent.purposeOfEvent. For example, during a machine-to-machine transfer it might not be obvious to the audit system who caused the event, but it does know why.
AuditEvent.source
Definition | The system that is reporting the event.
Control | 1..1
Requirements | The event is reported by one source.

Comments | Since multi-tier, distributed, or composite applications make source identification ambiguous, this collection of fields may repeat for each application or process actively involved in the event. For example, multiple value-sets can identify participating web servers, application processes, and database server threads in an n-tier distributed application. Passive event participants (e.g. low-level network transports) need not be identified.
AuditEvent.source.site
Definition | Logical source location within the healthcare enterprise network. For example, a hospital or other provider location within a multi-entity provider group.
Control | 0..1
Type | string
Requirements | This value differentiates among the sites in a multi-site enterprise health information system.
AuditEvent.source.identifier
Definition | Identifier of the source where the event was detected.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 1..1
Type | Identifier
Requirements | This field ties the event to a specific source system. It may be used to group events for analysis according to where the event was detected.
Summary | true
AuditEvent.source.type
Definition | Code specifying the type of source where event originated.
Control | 0..*
Terminology Binding | Audit Event Source Type (Extensible)
Type | Coding
Requirements | This field indicates which type of source is identified by the Audit Source ID. It is an optional value that may be used to group events for analysis according to the type of source where the event occurred.
AuditEvent.entity
Definition | Specific instances of data or objects that have been accessed.
Control | 0..*
Requirements | The event may have other entities involved.
Alternate Names | ParticipantObject
Comments | Required unless the values for event identification, agent identification, and audit source identification are sufficient to document the entire auditable event. Because events may have more than one entity, this group can be a repeating set of values.
Invariants | Defined on this element sev-1: Either a name or a query (NOT both) (expression : name.empty() or query.empty(), xpath: not(exists(f:name)) or not(exists(f:query)))
AuditEvent.entity.identifier
Definition | Identifies a specific instance of the entity. The reference should always be version specific.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
Comments | Identifier detail depends on entity type.
AuditEvent.entity.reference
Definition | Identifies a specific instance of the entity. The reference should be version specific.
Control | 0..1
Type | Reference(Any)
Summary | true
AuditEvent.entity.type
Definition | The type of the object that was involved in this audit event.
Control | 0..1
Terminology Binding | AuditEventEntityType (Extensible)
Type | Coding
Requirements | To describe the object being acted upon. In addition to queries on the subject of the action in an auditable event, it is also important to be able to query on the object type for the action.
Comments | This value is distinct from the user's role or any user relationship to the entity.
AuditEvent.entity.role
Definition | Code representing the role the entity played in the event being audited.
Control | 0..1
Terminology Binding | AuditEventEntityRole (Extensible)
Type | Coding
Requirements | For some detailed audit analysis it may be necessary to indicate a more granular type of entity, based on the application role it serves.
AuditEvent.entity.lifecycle
Definition | Identifier for the data life-cycle stage for the entity.
Control | 0..1
Terminology Binding | ObjectLifecycleEvents (Extensible)
Type | Coding
Requirements | Institutional policies for privacy and security may optionally fall under different accountability rules based on data life cycle. This provides a differentiating value for those cases.
Comments | This can be used to provide an audit trail for data, over time, as it passes through the system.
AuditEvent.entity.securityLabel
Definition | Security labels for the identified entity.
Control | 0..*
Terminology Binding | All Security Labels (Extensible)
Type | Coding

Requirements | This field identifies the security labels for a specific instance of an object, such as a patient, to detect/track privacy and security issues.
Comments | Copied from entity meta security tags.
AuditEvent.entity.name
Definition | A name of the entity in the audit event.
Control | 0..1
Type | string
Requirements | Use only where entity can't be identified with an identifier.
Summary | true
Comments | This field may be used in a query/report to identify audit events for a specific person. For example, where multiple synonymous entity identifiers (patient number, medical record number, encounter number, etc.) have been used.
Invariants | Affect this element sev-1: Either a name or a query (NOT both) (expression : name.empty() or query.empty(), xpath: not(exists(f:name)) or not(exists(f:query)))
AuditEvent.entity.description
Definition | Text that describes the entity in more detail.
Control | 0..1
Type | string
Requirements | Use only where entity can't be identified with an identifier.
AuditEvent.entity.query
Definition | The query parameters for a query-type entities.
Control | 0..1
Type | base64Binary
Requirements | For query events, it may be necessary to capture the actual query input to the query process in order to identify the specific event. Because of differences among query implementations and data encoding for them, this is a base 64 encoded data blob. It may be subsequently decoded or interpreted by downstream audit analysis processing.
Summary | true
Invariants | Affect this element sev-1: Either a name or a query (NOT both) (expression : name.empty() or query.empty(), xpath: not(exists(f:name)) or not(exists(f:query)))
AuditEvent.entity.detail
Definition | Tagged value pairs for conveying additional information about the entity.
Control | 0..*
Requirements | Implementation-defined data about specific details of the object accessed or used.
AuditEvent.entity.detail.type
Definition | The type of extra detail provided in the value.
Control | 1..1
Type | string
AuditEvent.entity.detail.value
Definition | The details, base64 encoded. Used to carry bulk information.
Control | 1..1
Type | base64Binary
Requirements | Should not duplicate the entity value unless absolutely necessary.
Comments | The value is base64 encoded to enable various encodings or binary content.

6.4.7 Resource AuditEvent - Mappings

URL: https://hl7.org/fhir/STU3/auditevent-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.4.7 Resource AuditEvent - Mappings

Security Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Mappings:

RIM Mapping

DICOM Tag Mapping

W3C PROV

FHIR Provenance

Mappings for the auditevent resource.

6.4.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

AuditEvent | ControlAct[moodCode=EVN]
type | .code (type, subtype and action are pre-coordinated or sent as translations)
subtype | .code (type, subtype and action are pre-coordinated or sent as translations)
action | .code (type, subtype and action are pre-coordinated or sent as translations)
recorded | .effectiveTime
outcome | .actionNegationInd
outcomeDesc | .outboundRelationship[typeCode=OUT].target.text
purposeOfEvent | * .reasonCode [ControlActReason when Act.class = CACT Control Act] *.outboundRelationship[typeCode=RSON].target
agent | .participation
role | .role and/or .typeCode and/or .functionCode

reference | .id
userId | .id
altId | .id (distinguish id type by root)
name | .name
requestor | If participation.typeCode was author, then true
location | * Role.Class =SDLOC *Role.Code = ServiceDeliveryLocationRoleType *Entity.Code = PlaceEntityType = df.Types of places for Entity.Class = PLC *EntityClass = PLC = df.A physical place or site with its containing structure. May be natural or man-made. The geographic position of a place may or may not be constant.
policy | ActPolicyType
media | .player.description.mediaType
network | .player.description.reference
address | pre-coordinated into URL
type | pre-coordinated into URL
purposeOfUse | *.reasonCode [ActHealthInformationPurposeOfUseReason codes/v:PurposeOfUse (2.16.840.1.113883.1.11.20448) * .outboundRelationship[typeCode=RSON or SUBJ].target
source | .participation[typeCode=INF].role[classCode=ASSIGN].player[classCode=DEV, determinerCode=INSTANCE]
site | .scopedRole[classCode=LOCE].player.desc
identifier | .id
type | .code
entity | .outboundRelationship[typeCode=SUBJ].target or .participation[typeCode=SBJ].role
identifier | .id.extension
reference | [self]
type | [self::Act].code or role.player.code
role | role.code (not sure what this would mean for an Act)
lifecycle | target of ObservationEvent[code="lifecycle"].value
securityLabel | .confidentialityCode
name | .title
description | .text
query | No mapping
detail | .inboundRelationship[typeCode=SUBJ].target[classCode=OBS, moodCode=EVN]
type | .code
value | .value

6.4.7.2 Mappings for DICOM Tag Mapping (http://nema.org/dicom)

AuditEvent | Message
type | EventId
subtype | EventTypeCode
action | EventActionCode
recorded | EventDateTime
outcome | EventOutcomeIndicator
outcomeDesc | EventOutcomeDescription
purposeOfEvent | EventPurposeOfUse
agent | ActiveParticipant
role | RoleIdCode
reference
userId | UserId
altId | AlternativeUserId
name | UserName
requestor | UserIsRequestor
location
policy | ParticipantRoleIDCode
media | MediaType
network
address | NetworkAccessPointID
type | NetworkAccessPointTypeCode
purposeOfUse
source | AuditSourceIdentification
site | AuditEnterpriseSiteId
identifier | AuditSourceId
type | AuditSourceTypeCode
entity | ParticipantObjectIdentification
identifier | ParticipantObjectID and ParticipantObjectIDTypeCode
reference | ParticipantObjectID
type | ParticipantObjectTypeCode
role | ParticipantObjectTypeCodeRole
lifecycle | ParticipantObjectDataLifeCycle
securityLabel | ParticipantObjectSensitivity
name | ParticipantObjectName
description | ParticipantObjectDescription
query | ParticipantObjectQuery
detail | ParticipantObjectDetail
type | ParticipantObjectDetail.type

value | ParticipantObjectDetail.value

6.4.7.3 Mappings for W3C PROV (http://www.w3.org/ns/prov)

The provenance resource is based on known practices in the HL7 implementation space, particularly those found in the v2 EVN segment, the v3 ControlAct Wrapper, the CDA header, and IHE ATNA. The conceptual model underlying the design is the W3C provenance Specification . Though the content and format of the resource is designed to meet specific requirements for FHIR, all the parts of the resource are formally mapped to the PROV-O specification, and FHIR resources can be transformed to their W3C PROV equivalent.

AuditEvent
type | Activity
subtype
action
recorded | Activity.when
outcome
outcomeDesc
purposeOfEvent | Activity.Activity
agent | Agent
role | Agent.Attribution
reference | Agent.Identity
userId | Agent.Identity
altId | Agent.Identity
name | Agent.Identity
requestor
location | Activity.location
policy
media
network
address | Agent.Location
type
purposeOfUse | Agent.Activity
source
site
identifier
type
entity | Entity
identifier
reference
type | Entity.type
role | Entity.role
lifecycle | Entity.role
securityLabel
name | Entity.Label
description
query
detail
type
value

6.4.7.4 Mappings for FHIR Provenance (http://hl7.org/fhir/provenance)

AuditEvent
type
subtype
action
recorded | Provenance.recorded
outcome
outcomeDesc
purposeOfEvent | Provenance.reason, Provenance.activity
agent | Provenance.agent
role | Provenance.agent.role
reference | Provenance.agent.actor
userId | Provenance.agent.userId
altId
name
requestor
location | Provenance.location
policy | Provenance.policy
media
network
address
type
purposeOfUse

source
site
identifier
type
entity | Provenance.target, Provenance.entity
identifier
reference | Provenance.target, Provenance.entity.reference
type | Provenance.entity.type
role
lifecycle | Provenance.entity.role
securityLabel
name
description
query
detail
type
value

6.4.8 Resource AuditEvent - Design Notes

URL: https://hl7.org/fhir/STU3/auditevent-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.4.8 Resource AuditEvent - Design Notes

to do

6.4.9 Resource AuditEvent - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/auditevent-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.4.9 Resource AuditEvent - Extensions & Profiles

Security Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the AuditEvent resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

6.4.9.1 Profiles

ehrsrle-auditevent | Defines the elements to be supported within the AuditEvent resource in order to conform with the Electronic Health Record System Functional Model Record Lifecycle Event standard | for EHRS FM Record Lifecycle Event - Audit Event

6.4.9.2 Extensions

auditevent-Accession | Accession | for AuditEvent HL7 Extensions
auditevent-Anonymized | Anonymized | for AuditEvent HL7 Extensions
auditevent-Encrypted | Encrypted | for AuditEvent HL7 Extensions
auditevent-Instance | Instance | for AuditEvent HL7 Extensions
auditevent-MPPS | MPPS | for AuditEvent HL7 Extensions
auditevent-NumberOfInstances | NumberOfInstances | for AuditEvent HL7 Extensions
auditevent-ParticipantObjectContainsStudy | ParticipantObjectContainsStudy | for AuditEvent HL7 Extensions
auditevent-SOPClass | SOPClass | for AuditEvent HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

6.4.9.3 Search Extensions

No Search Extensions defined for this resource

6.4.10 AuditEvent HL7 Extensions

URL: https://hl7.org/fhir/STU3/auditevent-extensions.html

Defines common extensions used with or related to the AuditEvent resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.4.10 AuditEvent HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

6.4.10.1 Content

Extensions:

auditevent-SOPClass | SOPClass : Required if ParticipantObjectIDTypeCode is (110180, DCM, "Study Instance UID") and any of the optional fields (AccessionNumber, ContainsMPPS, NumberOfInstances, ContainsSOPInstances,Encrypted,Anonymized) are present in this Participant Object. May be present if ParticipantObjectIDTypeCode is (110180, DCM, "Study Instance UID") even though none of the optional fields are present.

auditevent-Accession | Accession : An Accession Number associated with this participant object.

auditevent-MPPS | MPPS : An MPPS Instance UID associated with this entity.

auditevent-NumberOfInstances | NumberOfInstances : The Number of SOP Instances referred to by this entity.

auditevent-Instance | Instance : Th SOP Instance UID values.

auditevent-Encrypted | Encrypted : True or False indicating whether the data was encrypted.

auditevent-Anonymized | Anonymized : True or False indicating whether all patient identifying information was removed from the data.

auditevent-ParticipantObjectContainsStudy | ParticipantObjectContainsStudy : A Study Instance ID, which may be used when the Entity type is not (110180, DCM, "Study Instance UID").

6.4.11 Resource AuditEvent - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/auditevent-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

6.4.11 Resource AuditEvent - Detailed Descriptions

Security Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

R2 : R3 Converson maps for AuditEvent.

Functional status for this map: 8 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

6.4.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/AuditEvent2to3" = "R2 to R3 Conversions for AuditEvent"


uses "http://hl7.org/fhir/DSTU2/StructureDefinition/AuditEvent" alias  AuditEventR2 as source
uses "http://hl7.org/fhir/StructureDefinition/AuditEvent" alias  AuditEvent as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  AuditEvent extends DomainResource
  input src : AuditEventR2 as source
  input tgt : AuditEvent as target

  "AuditEvent-event" : for src.event as ev then {
     "AuditEvent-type" : for ev.type make tgt.type
     "AuditEvent-subtype" : for ev.subtype make tgt.subtype
     "AuditEvent-action" : for ev.action make tgt.action
     "AuditEvent-recorded" : for ev.dateTime make tgt.recorded
     "AuditEvent-outcome" : for ev.outcome make tgt.outcome
     "AuditEvent-outcomeDesc" : for ev.outcomeDesc make tgt.outcomeDesc
     "AuditEvent-purposeOfEvent" : for ev.purposeOfEvent make tgt.purposeOfEvent
  }
  "AuditEvent-agent" : for src.participant as vs0 make tgt.agent as vt0 then agent(vs0, vt0)
  "AuditEvent-source" : for src.source as vs0 make tgt.source as vt0 then source (vs0, vt0)
  "AuditEvent-entity" : for src.object as vs0 make tgt.entity as vt0 then entity (vs0, vt0)
endgroup

group agent extends BackboneElement
  input src as source
  input tgt as target

  "AuditEvent.agent-role" : for src.role make tgt.role
  "AuditEvent.agent-reference" : for src.reference make tgt.reference
  "AuditEvent.agent-userId" : for src.userId make tgt.userId
  "AuditEvent.agent-altId" : for src.altId make tgt.altId
  "AuditEvent.agent-name" : for src.name make tgt.name
  "AuditEvent.agent-requestor" : for src.requestor make tgt.requestor
  "AuditEvent.agent-location" : for src.location make tgt.location
  "AuditEvent.agent-policy" : for src.policy make tgt.policy
  "AuditEvent.agent-media" : for src.media make tgt.media
  "AuditEvent.agent-network" : for src.network as vs make tgt.network as vt then network(vs, vt)
  "AuditEvent-purposeOfUse" : for src.purposeOfUse make tgt.purposeOfUse
endgroup

group network extends BackboneElement
  input src as source
  input tgt as target
  "AuditEvent.agent.network-address" : for src.address make tgt.address
  "AuditEvent.agent.network-type" : for src.type make tgt.type
endgroup

group source extends BackboneElement
  input src as source
```

```
  input tgt as target
  "AuditEvent.source-site" : for src.site make tgt.site
  "AuditEvent.source-identifier" : for src.identifier make tgt.identifier
  "AuditEvent.source-type" : for src.type make tgt.type
endgroup

group entity extends BackboneElement
  input src as source
  input tgt as target
  "AuditEvent.entity-identifier" : for src.identifier make tgt.identifier
  "AuditEvent.entity-reference" : for src.reference make tgt.reference
  "AuditEvent.entity-type" : for src.type make tgt.type
  "AuditEvent.entity-role" : for src.role make tgt.role
  "AuditEvent.entity-lifecycle" : for src.lifecycle make tgt.lifecycle
  "AuditEvent.entity-securityLabel" : for src.securityLabel make tgt.securityLabel
  "AuditEvent.entity-name" : for src.name make tgt.name
  "AuditEvent.entity-description" : for src.description make tgt.description
  "AuditEvent.entity-query" : for src.query make tgt.query
  "AuditEvent.entity-detail" : for src.detail as vs make tgt.detail as vt then detail(vs, vt)
endgroup

group detail extends BackboneElement
  input src as source
  input tgt as target
  "AuditEvent.entity.detail-type" : for src.type make tgt.type
  "AuditEvent.entity.value-type" : for src.value make tgt.value
endgroup
```

6.4.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/AuditEvent3to2" = "R3 to R2 Conversions for AuditEvent"

uses "http://hl7.org/fhir/StructureDefinition/AuditEvent" alias  AuditEvent as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/AuditEvent" alias  AuditEventR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types AuditEvent extends DomainResource
  input src : AuditEvent as source
  input tgt : AuditEventR2 as target

  "AuditEvent-event" : for src make tgt.event as ev then {
     "AuditEvent-type" : for src.type make ev.type
     "AuditEvent-subtype" : for src.subtype make ev.subtype
     "AuditEvent-action" : for src.action make ev.action
     "AuditEvent-recorded" : for src.recorded make ev.dateTime
     "AuditEvent-outcome" : for src.outcome make ev.outcome
     "AuditEvent-outcomeDesc" : for src.outcomeDesc make ev.outcomeDesc
     "AuditEvent-purposeOfEvent" : for src.purposeOfEvent make ev.purposeOfEvent
  }
  "AuditEvent-agent" : for src.agent as vs0 make tgt.participant as vt0 then agent(vs0, vt0)
  "AuditEvent-source" : for src.source as vs0 make tgt.source as vt0 then source (vs0, vt0)
  "AuditEvent-entity" : for src.entity as vs0 make tgt.object as vt0 then entity (vs0, vt0)
endgroup

group agent extends BackboneElement
  input src as source
  input tgt as target

  "AuditEvent.agent-role" : for src.role make tgt.role
  "AuditEvent.agent-reference" : for src.reference make tgt.reference
  "AuditEvent.agent-userId" : for src.userId make tgt.userId
  "AuditEvent.agent-altId" : for src.altId make tgt.altId
  "AuditEvent.agent-name" : for src.name make tgt.name
  "AuditEvent.agent-requestor" : for src.requestor make tgt.requestor
  "AuditEvent.agent-location" : for src.location make tgt.location
  "AuditEvent.agent-policy" : for src.policy make tgt.policy
  "AuditEvent.agent-media" : for src.media make tgt.media
  "AuditEvent.agent-network" : for src.network as vs make tgt.network as vt then network(vs, vt)
  "AuditEvent-purposeOfUse" : for src.purposeOfUse make tgt.purposeOfUse
endgroup

group network extends BackboneElement
  input src as source
  input tgt as target
  "AuditEvent.agent.network-address" : for src.address make tgt.address
  "AuditEvent.agent.network-type" : for src.type make tgt.type
endgroup

group source extends BackboneElement
  input src as source
  input tgt as target
  "AuditEvent.source-site" : for src.site make tgt.site
  "AuditEvent.source-identifier" : for src.identifier make tgt.identifier
  "AuditEvent.source-type" : for src.type make tgt.type
endgroup

group entity extends BackboneElement
  input src as source
  input tgt as target
```

```
  "AuditEvent.entity-identifier" : for src.identifier make tgt.identifier
  "AuditEvent.entity-reference" : for src.reference make tgt.reference
  "AuditEvent.entity-type" : for src.type make tgt.type
  "AuditEvent.entity-role" : for src.role make tgt.role
  "AuditEvent.entity-lifecycle" : for src.lifecycle make tgt.lifecycle
  "AuditEvent.entity-securityLabel" : for src.securityLabel make tgt.securityLabel
  "AuditEvent.entity-name" : for src.name make tgt.name
  "AuditEvent.entity-description" : for src.description make tgt.description
  "AuditEvent.entity-query" : for src.query make tgt.query
  "AuditEvent.entity-detail" : for src.detail as vs make tgt.detail as vt then detail(vs, vt)
endgroup

group detail extends BackboneElement
  input src as source
  input tgt as target
  "AuditEvent.entity.detail-type" : for src.type make tgt.type
  "AuditEvent.entity.value-type" : for src.value make tgt.value
endgroup
```

6.4.11.3 R3 Validation Errors